



Final Report

Preparing to study the surface of Phobos using IDEFIX WheelCam images

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Abstract

The Martian Moons eXploration (MMX) mission, led by JAXA and scheduled for launch in 2026, represents a pivotal opportunity to advance our understanding of Mars' natural satellites, Phobos and Deimos. This research supports MMX's scientific goals by investigating the feasibility of using WheelCam imagery from the IDEFIX rover to analyze Phobos' surface properties. Two primary objectives are addressed: (1) estimating wheel sinkage from imagery as a means of determining regolith mechanical properties, and (2) evaluating the potential for spectral analysis using panchromatic images under narrow-band LED illumination. The sinkage analysis method demonstrated robust wheel contour detection across a range of regolith environments. However, performance was strongly influenced by illumination conditions, with optimal image saturation levels varying as a function of surface reflectivity. Initial validation of the sinkage estimates yielded an absolute error below 5% relative to laser measurements, although uncertainties in the reference measurements and the limited validation dataset indicate that more experiments are needed to quantify the pipeline's accuracy and reliability. Spectral studies demonstrated meaningful correlations between image brightness and reflectance, validating the approach under controlled conditions. A custom testbed simulating Phobos-like conditions enabled robust method validation. The research highlights key next steps, including validation with geological analogs, calibration refinement, and algorithm optimization, to ensure reliable scientific return from the IDEFIX rover's imaging systems.

Keywords: Phobos, MMX, regolith characterization, planetary science, image processing, wheel sinkage, spectral imaging

Signed Declaration of Authenticity

This assignment is entirely our own work. Quotations from literature are properly indicated with appropriate references in the text. All literature used in this piece of work is indicated in the bibliography placed at the end. We confirm that no sources have been used other than those stated.

We understand that plagiarism (copy without mentioning the reference) is a serious examination offence that may result in disciplinary action being taken.

Date: April 6, 2026

Signature

Florian Mellab

A handwritten signature in black ink, appearing to read 'Florian Mellab', with a long horizontal stroke extending to the left.

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Introduction

Over the past century, humanity’s exploration of the Solar System has made extraordinary advances. Despite this progress, many regions of space remain unstudied and undiscovered due to the limitations of our technology. The MMX mission (Martian Moons eXploration) is the first of its kind, in which the close observation of the unexplored Martian moons will become possible. The moons of Mars, named Phobos and Deimos, were first discovered in 1877 by Asaph Hall [1] and their origin remains a debate up to this day, with three hypotheses that are all feasible under the current available data. The origin of the moons is believed to either result from a giant impact, the capture of asteroids or co-formation with Mars [2, 3].

Other missions in the past, especially by the Soviet Union and Russian, have left Earth to explore the moons of Mars and failed due to technical difficulties associated with long-distance spaceflight or environmental reasons related to the size, orbit and gravity field of these objectives. Now, set to depart in 2026, JAXA’s (Japan Aerospace Agency) MMX is designed with experience from the past missions and with mechanisms to overcome previous failure. These include redundant hardware associated to robust software validation, advanced GNC system to track Phobos’ particular orbit, and specific landing systems and procedure for the weak gravity on this moon [4, 5].

The MMX mission aims to collect invaluable images and data from Phobos and Deimos, as well as collect samples that will be taken back to Earth for their study. Touch-and-go procedures, used in the past for sample-return missions, are not feasible due to engineering constraints [6]. Therefore, the MMX spacecraft will land on the surface of Phobos, the larger of the two moons, to take the samples. The surface conditions of Phobos are unknown and this poses a risk for landing. Consequently, it will first drop the IDEFIX rover [5] from approximately 100 m to assess the risks on the ground. Through the IDEFIX rover and its different instruments, the surface properties of Phobos will be studied.

The focus of this project lies on determining the possibility to study the mechanical and mineralogical properties of the surface of Phobos through the rover’s WheelCams, a set of panchromatic cameras that point to each of the wheels of the rover [7]. More specifically, this report will study the feasibility of gathering relevant information through the observation of the wheel sinkage into the surface of Phobos and the use of different coloured LEDs to obtain a spectral study of the surface. Both of these approaches remain in the initial stages and in this report, the advancements will be discussed, as well as the possible limitations of the procedures and conclusions drawn from the testing made throughout the project. Finally, the future of this investigation will be discussed, including the next steps and objectives.

1 Background and Objectives

In this chapter, the scientific background of our project will be presented. section 1.1 will deal with the general context of the MMX mission and how it relates to the goals that will be proposed for this research project. section 1.2 will focus on the theoretical background, explaining the conceptual basis needed for the completion of the project as well as how this research will contribute to the state of the art in this domain. Finally, in section 1.3, the general objectives of the research will be discussed as well as the objectives specific to this semester.

1.1 Context and key issues

1.1.1 Mission context

This research project focuses on the Martian Moons eXploration (MMX) mission [6], led by the Japan Aerospace Exploration Agency (JAXA) in collaboration with Centre National d'Etudes Spatiales (CNES) and the Deutsches Zentrum für Luft- und Raumfahrt (DLR). MMX will be the first sample-return mission from Phobos, the larger of Mars' two moons, and it will also conduct the most detailed investigation to date of both Phobos and Deimos.

Although past missions such as NASA's Viking program have gathered detailed images of the Martian moons [8], and others like Phobos 1, Phobos 2, and Phobos-Grunt have attempted direct investigation [9], only Phobos 2 succeeded in transmitting scientific data. These limited and incomplete missions have left many unanswered questions about the origin, composition and evolution of Mars' moons.

This knowledge gap has led to the proposal of several mission concepts over the years, but MMX remains the only currently approved mission to investigate the Martian moons [6]. Initially proposed in 2015, MMX is scheduled to launch in 2026.

After launch, the spacecraft will head towards Mars where, upon arrival, it will enter orbit around the planet. Afterward, it will transition into orbit around Phobos, where it will eventually land to collect surface material. Once the samples have been collected and the spacecraft commences its return journey, MMX will perform a close flyby of Deimos, to obtain high resolution images and gather spectral data.

Although Phobos remains unfamiliar in many aspects, some of its general characteristics have been established through terrestrial observations and previous Mars missions. These include estimates on its size, mass, orbit, surface gravity, and overall shape [10]. As a result, the design of the MMX spacecraft can begin with a set of relatively well constrained physical parameters of its target, unlike previous asteroid missions.

The main objectives of the MMX mission are focused on revealing the origins of Mars' moons by the study of their composition through both close-range observations and the return of physical samples. Due to the limited knowledge about the physical and mechanical properties of Phobos' surface, the safe landing to collect a sample poses significant challenges. To reduce this risk, the

mission includes IDEFIX, a small French-German rover designed to land on Phobos before the main spacecraft. IDEFIX rover has four main scientific instruments to obtain measurements from the surface of Phobos [5]. These include two navigational cameras (NavCams) [11], two cameras pointing towards the wheels (WheelCams) [7], a Raman spectrometer known as RAX [12] and a miniature radiometer called miniRAD [13]. Each instrument serves a purpose and together they provide complementary data that will allow the rover to conduct in-situ measurements to evaluate the surface conditions, determining whether it is safe for landing to occur and for the sample to be taken. Moreover, it will gather valuable information on the moon's surface characteristics, which will contribute to the overall scientific goals of the mission.

The rover is expected to be released from an altitude of less than 100 meters (ideally around 45 meters), descending freely without any braking mechanism until it impacts the surface, where it will right itself by repositioning its legs to achieve a stable orientation. The overall fall velocity will be low, as the gravitational acceleration on Phobos is about 0.0057 m/s^2 . Once upright, the rover will deploy its solar panels and begin its surface operations. However, the movement speed and the total area covered is unknown as it is constrained to the terrain conditions [5]. Although the mission aims for the rover to operate for approximately 100 days, actual performance will depend on surface conditions [14].

1.1.2 Mars' moons: Phobos

There are several competing theories regarding the origin of the Martian moons, however their exact formation remains unknown. Due to their shape and size, it is believed that they could be captured asteroids. Another theory suggests that they are the result of debris that had its origin on a giant collision between Mars and a protoplanet [15]. A third one has proposed that they are a result of co-formation, formed at the same time as Mars, however, the low density of the moons makes this theory the least probable of the three [16].

Phobos is the larger and innermost of Mars' two moons, orbiting at a gradually diminishing distance of approximately 6000 kilometers from the Martian surface. Due to the gravitational pull of Mars, Phobos' orbit is slowly decaying. Therefore, the moon is expected to either crash into the planet or break up into a ring in approximately 50 million years [17].

As mentioned in subsection 1.1.1, the general characteristics of Phobos are established with a certain degree of confidence. Its overall shape is very irregular due to the presence of several craters, and its mass is insufficient to make it a spherical body.

Regarding the composition of the surface, constraints exist, although with a high degree of ambiguity. The surface of Phobos has been studied through spectral imaging, and related to others of known composition and similar spectral characteristics [18]. One of the most distinct characteristics of the surface of Phobos is the fact that it exhibits two clear geological units, red and blue, of which the origin is unknown, very close to its main crater Stickney [19, 18]. The spectral characteristics of these geological units have been studied and have proven to exhibit similarities; its origin is also a mystery that the MMX mission will address [6, 14].

In general, Phobos presents a very dark spectra and very low albedo, comparable to P-type and D-type asteroids [18], which are dark, primitive and carbon-rich bodies with a reddish

spectra. Nevertheless, its spectral characteristics are compatible still with the both of principal formation theories [15].

Among the minerals of particular interest in this context are olivine and pyroxene. These silicate minerals are commonly found in basaltic materials and are major components of planetary crusts and mantles. Their presence on Phobos would therefore provide important clues about its origin. In particular, the detection of olivine and pyroxene would support a formation scenario linked to material originating from Mars, as these minerals are abundant in Martian rocks. On the other hand, a lack of these minerals could be more consistent with a primitive asteroid origin. For this reason, studying the spectral response of olivine and pyroxene samples is highly relevant, as it will allow us to assess whether the experimental method is capable of identifying materials that are key to distinguishing between the different formation hypotheses of Phobos.

The surface of the moon is known to be covered by regolith, which is most likely formed by the accumulation of impact debris on its surface [19]. The properties of the regolith evolve over the history of the body and can differ depending on the mineralogy present on Phobos, impacts, thermal alterations and fatigue and space weathering. Space weathering is a process in which a body will darken or redden due to the pass of time, deepening the absorption bands in the optical spectra. It is the explanation for the differences that occur on bodies of seemingly similar composition but with different spectral properties [20].

Other properties, such as mechanical properties of the regolith are related to the granular behavior of the regolith and particle size. For these properties, theoretical estimations have been made due to limited knowledge on how granular materials behave on low gravity environments. The range of particle size is believed to be between 30 μm to 10 cm [10], however, it should be noted that given the concerns about the landing site, the images taken by IDEFIX will likely be biased towards finer-grained surface materials [7].

1.1.3 IDEFIX Rover: WheelCams

This research project focuses mainly on the WheelCams. These cameras will observe the interaction between the wheels and the regolith, obtaining crucial information that will help determine the properties of the regolith. Their field of view is represented in red on Figure 1.1.

The WheelCams are based on a multi-purpose camera for Space exploration named CASPEX, which was developed by CNES in 2015 [7]. The WheelCams have a bandwidth of 550 to 900nm, a field of view of 32.5° and take images of size 2048x2048 pixels. Each WheelCam is also be equipped with 7 LEDs, 4 White LEDs used for navigating and 3 colored LEDs (590 nm, 720 nm and 880 nm) to illuminate the environment [7].

The WheelCams are panchromatic cameras, in other words, they will receive information inside of a specific spectral range (corresponding to their bandwidth) and they will join said signal into a single output, a grayscale image. The combination of these images with the LEDs will allow to obtain spectral information that will assist into the compliance of the mission objectives.

The specific objectives from the IDEFIX WheelCams are the following [7]:

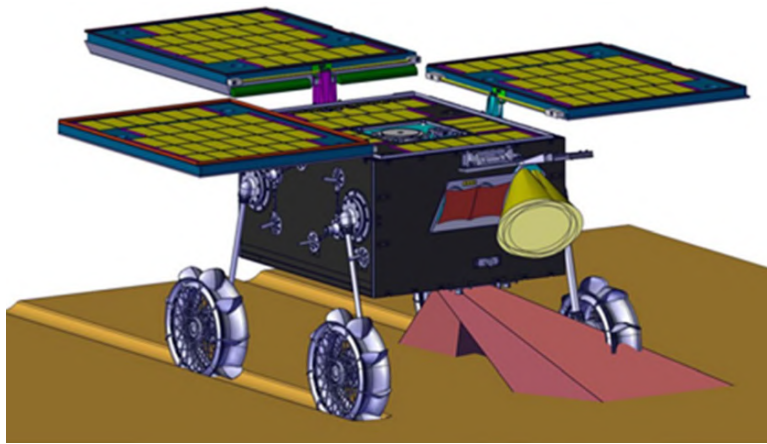


Figure 1.1: IDEFIX rover with deployed wheels and solar panels [5]

1. Determine the physical properties of the regolith particles
2. Determine the bulk mechanical properties of the regolith
3. Determine the dynamical behavior of the regolith
4. Observe possible layering in the shallow sub-surface
5. Constrain the mineralogical composition of the surface material
6. Assess space weathering
7. Determine regolith geological classes
8. Constrain the absolute local gravitational acceleration

Once again, the completion of these objectives will be possible by studying the interaction between the wheel and the regolith and by evaluating the information obtained by the use of the LEDs. In order to comprehend the possible conditions of the regolith on Phobos as well as to study the feasibility of performing certain analysis, a testbed was constructed to recreate such an environment and to generate sample images representative to those the WheelCam will acquire on Phobos.

1.1.4 Experimental testbed

The main objective of the testbed is to generate sample images which will allow to develop the image processing tools to prepare for the exploration of Phobos by the IDEFIX rover [7]. The testbed, shown in Figure 1.2, consists of three main sub-assemblies (the main structure, the material container, and the wheel assembly), multiple sensors (a motor encoder, two displacement sensors, and a laser profilometer), and two cameras. The two cameras are positioned to observe the wheel-regolith interactions from the same perspective as the front and rear IDEFIX WheelCams respectively. The cameras featured in the testbed are display similar field of views to those of the WheelCams installed on IDEFIX.

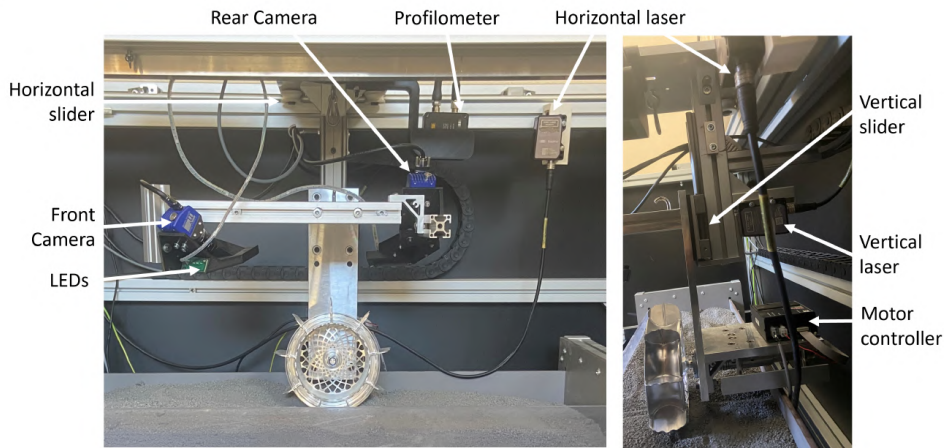


Figure 1.2: ISAE-SUPAERO WheelCam testbed [7]

The testbed is an important tool in this research project to acquire images with which the image processing algorithms will be developed and to test the algorithm in a range of conditions expected on the Martian moon. In particular, the regolith simulant in the testbed can be changed to simulate different conditions.

Three types of simulant were used to test the vision-based analysis pipeline. The JAXA simulant, designed to replicate Phobos regolith as accurately as possible, was used for initial development of the algorithm. The pipeline was then tested using Lunar regolith simulant (later labelled Moon simulant) that is composed of a fine dust with coarser grains. Finally, a gravel simulant with much coarser and larger particles was studied. Both the image processing algorithm and the results for these different simulant material are discussed in chapter 2.

The testbed, just as the IDEFIX rover, is also equipped with several LEDs for each camera. 2 white LEDs that will be used when the rover is in motion and 3 colored LEDs (with wavelengths of 590nm, 720nm and 880nm respectively) that will be used when the rover is static. Through the use of these LEDs and the panchromatic cameras, spectral studies will take place. The testbed will be used in chapter 3 to perform experiments to understand if and how spectral information about the regolith can be extracted from the WheelCam images by comparing to known spectral curves from the samples tested, discussed in further detail on subsection 3.4.3.

For both the sinkage estimation analysis and the spectral study, the images from the testbed constitute the main input on which development, testing and validation were carried out.

During the semester, it was identified by the engineering team that the initial testbed configuration did not accurately reproduce the viewing geometry of the IDEFIX rover cameras. Additionally, a misalignment of the LEDs was detected. As a result, the testbed was updated to better match the instrument configuration, including modifications to the camera optics and viewing angle, as well as a proper alignment of the LEDs.

The impact of these modifications can be observed in Figure 1.7, where a comparison between the before and after images is presented.



Figure 1.3: Front WheelCam in the previous configuration

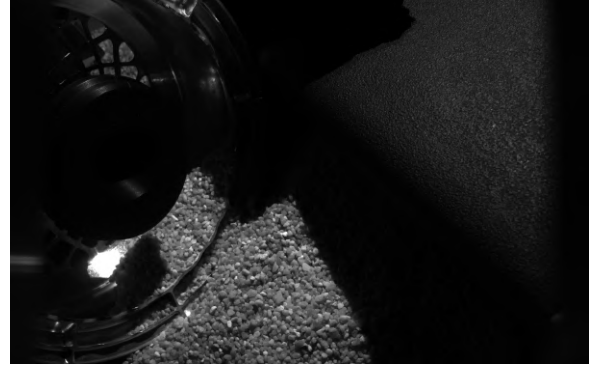


Figure 1.4: Front WheelCam in the updated configuration

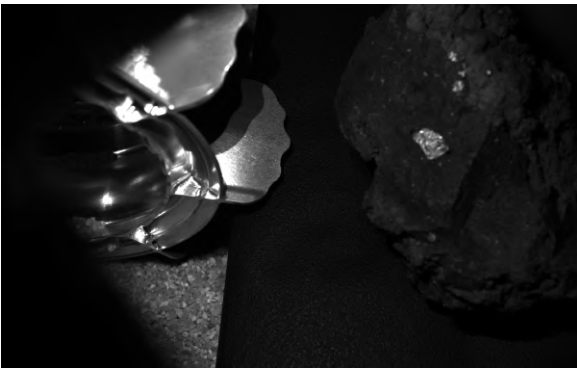


Figure 1.5: Rear WheelCam in the previous configuration

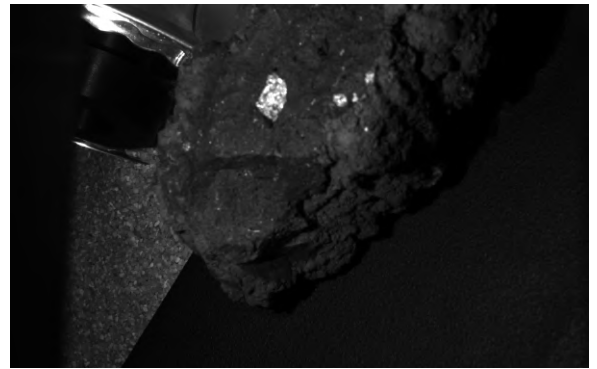


Figure 1.6: Front WheelCam in the updated configuration

Figure 1.7: WheelCams Configuration Update Example with analogue sample.

The impact of this readjustment on the obtained results will be discussed in section 2.4 and section 3.5.

1.2 State of the Art and novelty of this research

The state of the art in performing science with the WheelCam images is advancing on many fronts. Image processing algorithms are currently being developed at ISAE-SUPAERO to support characterization activities in preparation for receiving the first images from the IDEFIX rover on Phobos, aiming to provide insights into each of the WheelCam science objectives presented in subsection 1.1.3. Some examples of the analyses being developed include studies on the trench morphology left by the rover's tracks, which hold information about regolith cohesion. Work is also being done to study the particle size frequency distribution (SFD) of the regolith, which will reveal information about processes such as impacting and transporting experienced by the surface material [7].

In terms of studying sinkage, students at ISAE-SUPAERO have already conducted research to develop an image processing algorithm to detect the wheel and estimate sinkage [21]. Their work

also incorporated a homography transformation to correct perspective distortions and recover a perpendicular view of the wheel, as well as an ellipse-fitting algorithm to model the wheel contour. Building on this foundation, the present study introduces a novel approach based on classical computer vision techniques rather than ML, extends the analysis to both rear and front wheels, and investigates the impact of illumination conditions on algorithm performance. Additionally, it integrates experimental validation using a vertical laser setup on the testbed with the aim of providing a robust framework for automatic sinkage estimation.

On the topic of performing spectral analysis using the WheelCam images, no prior research has been carried out specifically on this issue. However, there is evidence from literature of similar research being carried out. In particular, studies have been done on reconstructing hyperspectral images using LED illumination in narrow wavelengths from a monochromatic camera [22]. Their research demonstrates that spectral information can be extracted from a series of monochromatic images illuminated by RGB LEDs of different wavelengths [22]. In contrast, the work discussed in this report focuses primarily on the identification of materials from measured reflectance, rather than on reconstructing spectral information. A key difficulty of this approach lies in the limited spectral sampling, as only three LEDs are available for comparison.

1.3 Aims and Objectives

The goal of this research project is to prepare for the scientific analysis of WheelCam images from Phobos by developing automated, reproducible tools capable of extracting wheel sinkage and spectral information. These tools must be robust and feasible considering the time and operational constraints of the MMX mission. To ensure reliability, the methods will be tested across a broad range of simulated conditions that mimic the expected environment on Phobos.

The specific objectives for this research are:

- To design an algorithm that can accurately identify the wheel, detect the regolith–wheel interface, and quantify the depth of sinkage.
- To assess whether the material composition of the regolith can be determined from the response of the regolith to illumination by LEDs at specific wavelengths.

These objectives directly support the WheelCam science goals, as outlined in subsection 1.1.3:

- **Sinkage estimation:** contributes to science objectives 1 (determining physical properties of regolith particles) and 2 (determining bulk mechanical properties of regolith).
- **Spectral analysis:** contributes to science objectives 5 (constraining mineralogical composition) and 6 (assessing space weathering).

2 Wheel Sinkage

2.1 Introduction

Wheel sinkage, defined as the vertical displacement of a wheel as it penetrates the soil, can provide insight into various physical properties of planetary regolith. Indeed, wheel sinkage is related to the soil's bearing capacity (ie. the maximum pressure it can withstand before undergoing shear failure) and regolith mechanical characteristics including cohesion and friction angle[7]. These properties are important to quantify as they influence the rover's locomotion and the landing dynamics for the main MMX spacecraft, which will land on Phobos in the weeks following the beginning of IDEFIX's exploration. Thus, this chapter focuses on developing a method to estimate sinkage from WheelCam images, issued from an experimental testbed. These results, along with other properties studied under the WheelCam science objectives, are key inputs for the interpretation of mechanical properties, and will contribute to determining the geological classification of the regolith[7].

2.2 Theoretical Background

Measurements of wheel sinkage and slip can yield valuable insights into the mechanical properties of the regolith. However, since the IDEFIX rover does not provide telemetry on motor torque or current, these parameters will be derived exclusively from WheelCam imagery [7].

A modeling framework must be established to link observed wheel behavior to the underlying physical and mechanical characteristics of Phobos' regolith. Fitting semi-empirical terramechanics models (such as the Bekker or Reece-Wong models) to the observed sinkage and slip data yields reliable estimations of key mechanical properties of the regolith, including cohesion, internal friction angle, shear strength, and stiffness [7].

However, these models must be adapted to the low-gravity environment on Phobos to provide accurate results. Low gravity reduces the normal force acting on the wheels and affects the behavior of the regolith as it interacts with the wheels. As a result, significantly higher wheel slippage is expected to occur in this type of environment and model parameters must be adjusted accordingly[7].

The bearing capacity of the regolith, a property crucial for predicting spacecraft landing dynamics, will be estimated independently using both direct sinkage measurements and analysis of the trench morphology left by the rover's tracks [7].

2.3 Methodology

This section describes the wheel detection and sinkage estimation pipeline. The algorithm is first presented for rear WheelCam images and then extended to front WheelCam images with

additional processing steps.

The pipeline consists of image segmentation, contour detection, ellipse fitting, and a homography transform to correct perspective distortion. Sinkage is then estimated from the intersection between the predicted wheel contour and the regolith surface. The different steps of this process and a snapshot of their results are summarised in Figure 2.1 and further detailed in the subsequent chapters. Most image processing operations were implemented using the OpenCV library in Python [23].

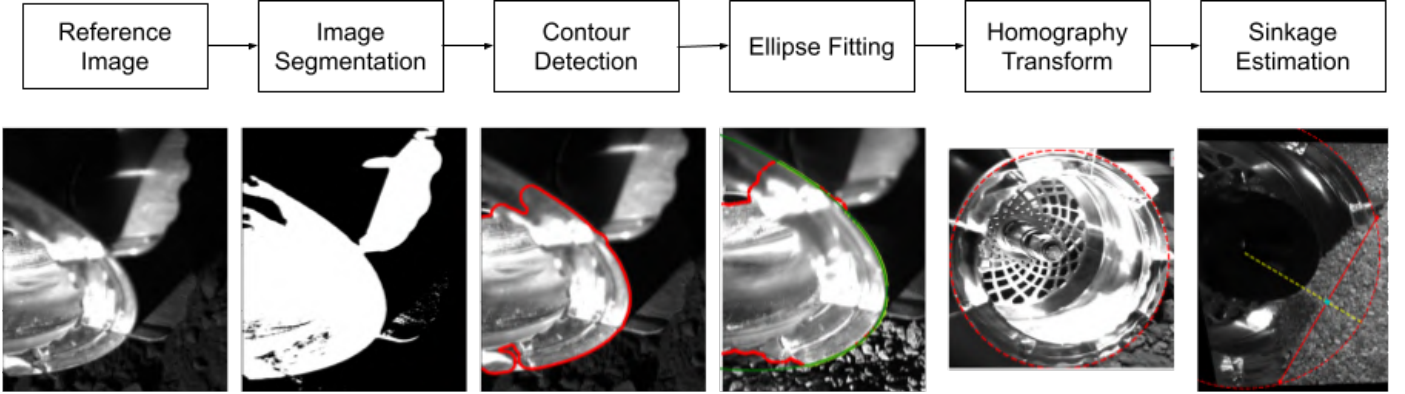
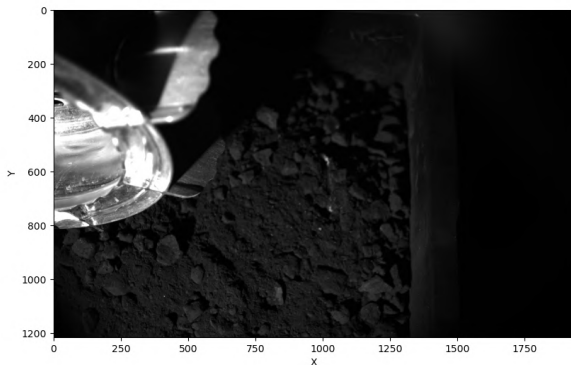


Figure 2.1: Steps of the WheelCam image processing pipeline

The images shown in Figure 2.2 taken using the testbed's rear and front WheelCams respectively with JAXA regolith simulant, and will be used throughout this chapter to illustrate the effect of the various processing steps in the pipeline. A thorough analysis of the algorithm's performance on unseen data will be important to assess the range of application of this algorithm, to ensure image processing operations run smoothly when the first images are acquired by IDEFIX on Phobos.



(a) Reference testbed rear WheelCam image with JAXA regolith simulant



(b) Reference testbed front WheelCam image with JAXA regolith simulant

Figure 2.2: Testbed rear and front WheelCam reference images

2.3.1 Image segmentation

This subsection focusses on image segmentation, a process consisting in differentiating and dividing the image's pixels into distinct groups to simplify its analysis. The pipeline for the image segmentation part of the wheel detection algorithm will be broken down, highlighting the main processing steps and the key decisions that shaped the algorithm.

Segmentation threshold

The definition of a threshold for binary segmentation is a key step in the algorithm. For this application, global thresholding, using a single intensity value as division threshold, was chosen instead of local (adaptive) thresholding, where the limit intensity of a region is dependent on that of the surrounding pixels and differ across the image. This is because the illumination intensity is nearly constant across the image, except for a bright object in the foreground.

One common method for global thresholding is Otsu thresholding [24], which determines an optimal threshold by maximizing the variance between the two pixel classes (interclass variance), that are pixels belonging to the same group post-segmentation - i.e. the wheel and the background. The advantage of this method is that the threshold is computed directly from the image, requiring no additional parameters and keeping the processing pipeline deterministic and unbiased.

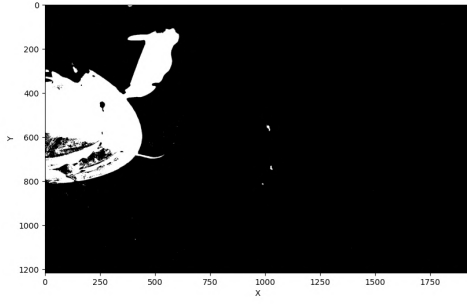
Another approach defines adaptive bounds based on the average pixel value within a predefined "wheel confidence area", a region where the wheel is expected to be present in all images [25]. In this work, a 200×200 pixel area located fully inside the wheel, on the image's left side, was used. The mean pixel value in this region is computed and the segmentation bounds are defined as $[P_a \cdot \text{Average}, P_d \cdot \text{Average}]$, where P_a and P_d are tuning parameters. Pixels with values within this range are set to 255 and all others to 0 to obtain a binary mask. For the reference image, the best performance was obtained with $P_a = 0.35$ and $P_d = 1.85$, through a trial and error method. These values provided a segmentation mask preserving most of the wheel while not including elements from the background.

Both global thresholding methods were implemented to compare their performance on a set of test images. An example of the segmented images obtained is shown in Figure 2.3. Visual inspection indicated that adaptive bounds performed better, as Otsu thresholding removed parts of the wheel edge and introduced more noise in the mask (Figure 2.3a) compared with the adaptive approach (Figure 2.3b).

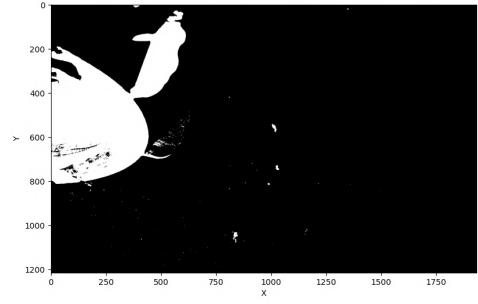
Morphological transformations

To improve the quality of the binary mask of the wheel and facilitate the fitting of an ellipse to the wheel edge at a later stage, morphological transformations should be applied. In particular, the protruding wheel grousers should be eliminated from the mask and the noise present in the foreground object should be removed as much as possible.

The two morphological transformations that were considered are OpenCV's opening and closing



(a) Binary segmentation using Otsu thresholding



(b) Binary segmentation using adaptive bounds with $P_a = 0.35$ and $P_d = 1.85$

Figure 2.3: Comparison of binary segmentation by Otsu thresholding and adaptive bounds

functions [26]. These operation rely on two key principles: erosion which redefine the intensity of a pixel to match all those under the kernel removing noise, and dilation which redefine the value of the pixel if any pixel under the kernel is white to fill gaps in the studied shape. The opening function allows to remove small protrusions from a main shape, see Figure 2.4a, and works on the principle of erosion followed by dilation. The closing function is used to remove noise from the foreground object, such as in Figure 2.4b, and works on the principle of dilation followed by erosion. For both of these operations, a key parameter to tune is the size and shape of the kernel, the structuring element.



(a) Example transformation using the opening function [26].



(b) Example transformation using the closing function [26].

Figure 2.4: Example images after morphological transformations

The choice of kernel allows to tune the transformations to act on objects of a certain dimension. Since the wheel grousers are large, an elliptical kernel of dimension (60, 60) was chosen for the opening function. The dimension of the noise elements being smaller, an elliptical kernel of dimension (6,6) was selected for the closing function. Note that the shape of the kernel is an ellipse inscribed within a rectangle of the specified size, so in this case the kernel is circular.

Through trial and error, the optimal order to perform the morphological operations is to apply one iteration of the closing function followed by one iteration of the opening function. The effects of these transformations on the binary mask are shown in Figure 2.5. As can be seen, the grouser has successfully been separated from the wheel, the noise within the mask has been reduced and the noise in the background has been eliminated completely. However, this comes

with a slight reduction in edge accuracy and the loss of some parts of the wheel.



(a) Binary segmented image before applying morphological transformations

(b) Segmented images after one iteration of closing and one iteration of opening

Figure 2.5: Morphological opening and closing operations applied to the segmented image

2.3.2 Contour detection

After the binary mask has been cleaned, edge detection is performed to extract the boundary of the mask. First, Gaussian blur is applied to the masks with a 5x5 square kernel. This operation averages the pixel's intensity within the kernel and smoothens the pixel intensity gradient. This operation is important to ensure the reliability and stability of the subsequent edge detection process, avoiding a 'staircased' boundary and eliminating false-edge detection from residual pixel-level variations. Then an edge map is obtained by performing Canny edge detection with sensitivity parameters $\text{minVal} = 220$ and $\text{maxVal} = 250$. OpenCV's Canny edge detector treats pixels whose value is above maxVal as "sure edges" and pixels below minVal as "sure non-edges" [27]. Pixels whose intensity lies between the two bounds are classified as edges only if one of the neighboring pixels is a "sure edge".

After generating the edge map, continuous contours are extracted using OpenCV's `findContours()` function. Each detected contour is then evaluated using the `minEnclosingCircle()` function, which computes the smallest circle capable of fully enclosing the contour [28]. Among these, the contour corresponding to the wheel is identified as the one whose enclosing circle has the largest radius. The continuous contour obtained for the reference image after the filtering process is shown in Figure 2.6.

2.3.3 Front WheelCam algorithm adjustments

The algorithm described so far was developed for rear WheelCam images. However, front WheelCam images are more suitable for sinkage estimation because the entire wheel and the wheel-regolith interface are visible. An adapted version of the algorithm was therefore developed for the front WheelCams, with additional processing steps and minor modifications to improve contour detection.

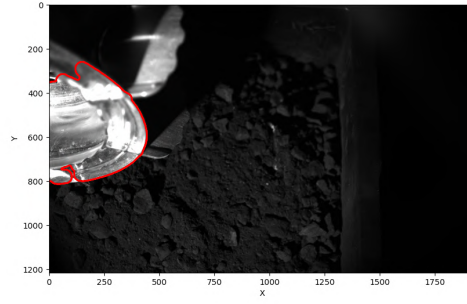


Figure 2.6: Contour selected after maximum radius filtering of their minimum enclosing circles

Preprocessing

An additional initial preprocessing step is required due to the uneven lighting across the front WheelCam image and a similar brightness between the wheel and its background making it more difficult to distinguish the wheel edge in some regions. This step consist in using OpenCV's Contrast Limited Adaptive Histogram Equalization (CLAHE) function, `cv2.createCLAHE`. This method enhances local image contrast by redistributing pixel intensities over a wider brightness range [29]. The reference front WheelCam image after CLAHE preprocessing is shown in Figure 2.7.

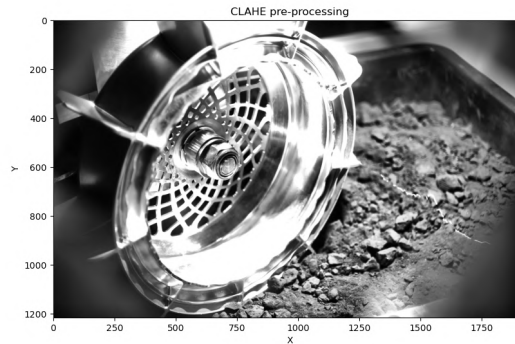


Figure 2.7: CLAHE preprocessing applied to front WheelCam reference image

Piecewise image segmentation

A common issue with front WheelCam images is that the wheel contour is detected primarily on the side where the contrast with the background is , here the left side of the image. To deal with this issue, the image is divided vertically at $x = 700$, and the segmentation step is performed independently on the left and right portions of the image. This allows the algorithm to better adapt its segmentation threshold for each side of the image, efficiently capturing the wheel interface.

This piecewise segmentation increases the likelihood of detecting contour points across the entire wheel boundary, which is needed to increase reliability of ellipse fitting as will be discussed in

subsection 2.3.4. The resulting contours are shown overlaid on the reference image in Figure 2.8.

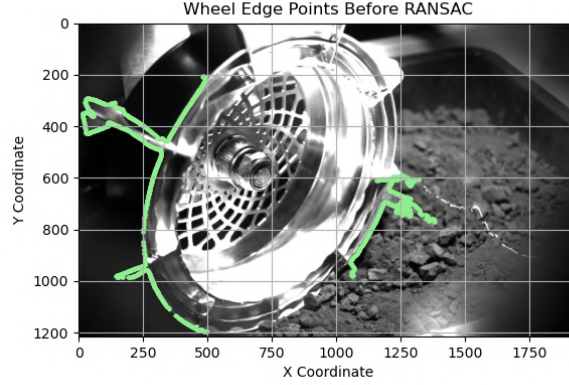


Figure 2.8: Piecewise segmentation contour patch front WheelCam

2.3.4 RANSAC based ellipse fitting

Due to the viewing angle of the camera, the circular wheel appears as an ellipse in the image plane. The next step in the algorithm pipeline is thus to fit an ellipse to the detected contour. However, the extracted contour is often noisy and may contain points that do not belong to the wheel boundary. One way to deal with outliers in a robust way is to use a Random Sampling Consensus (RANSAC) algorithm.

RANSAC is an iterative method that estimates model parameters while rejecting outliers. At each iteration, a random subset of contour points is selected and used to fit an ellipse model. The quality of the candidate model is then evaluated by computing the residual distance between the contour points and the ellipse. Points whose residuals fall below a predefined threshold are classified as inliers. The process is repeated for multiple random samples, and the model with the largest number of inliers is retained as the best estimate of the wheel contour.

The ellipse fitting procedure is implemented using the `skimage.measure.ransac` function together with the `EllipseModel`. The `ransac` function takes as input the data points and a model to be fitted, here the `EllipseModel`. This function runs the RANSAC algorithm for the ellipse model and outputs the final fitted ellipse model as well the inlier data points.

In this formulation, the ellipse is represented directly by five geometric parameters: the coordinates of the center (x_c, y_c) , the semi-major and semi-minor axes (a, b) , and the orientation angle θ of the ellipse relative to the image axes. A minimum of five points is required to estimate these parameters.

The number of RANSAC iterations required to obtain a reliable model depends on the proportion of inliers present in the data. Let w denote the fraction of inliers, n the minimum number of points required to fit the model ($n = 5$ for an ellipse), and p the desired probability of selecting at least one subset composed only of inliers. The required number of iterations k can be estimated by Equation 2.1 [30].

$$k = \frac{\log(1 - p)}{\log(1 - w^n)}. \quad (2.1)$$

The relationship between k and w for $p = 0.99$ and $n = 5$ is illustrated in Figure 2.9. As the proportion of outliers increases, the number of iterations required grows rapidly. This highlights the importance of obtaining a clean contour through the segmentation and morphological preprocessing steps described in subsection 2.3.1.

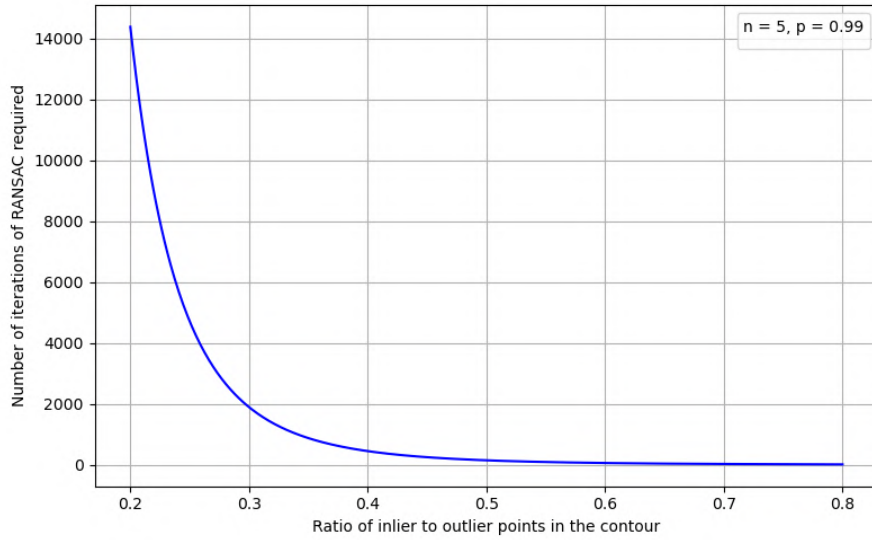
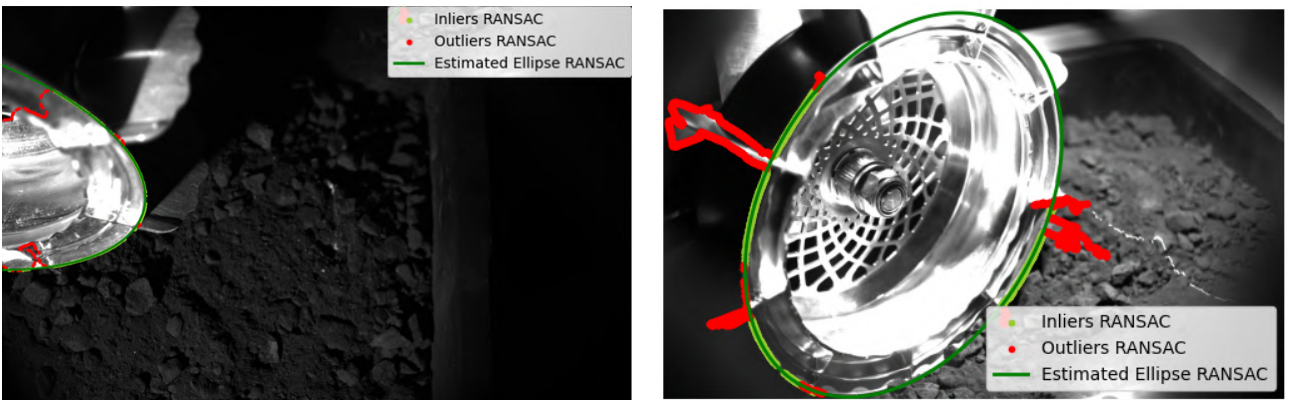


Figure 2.9: Number of RANSAC iterations required (k) as a function of the ratio of inliers to outliers (w) for $p=0.99$ and $n=5$

Examples of the resulting ellipse fits for the rear and front WheelCam reference images are shown in Figure 2.10.



(a) Rear WheelCam RANSAC ellipse fit

(b) Front WheelCam RANSAC ellipse fit

Figure 2.10: RANSAC ellipse fits for rear and front WheelCam reference images

2.3.5 Homography transform

Due to the angle of the camera, a circular wheel appears as an ellipse in the image plane. To recover a geometrically accurate representation of the wheel, a homography transformation is applied to map the detected ellipse back to a circle.

To compute the homography matrix, points are first sampled on the predicted ellipse.

$$\begin{aligned}x(t) &= x_c + a \cos(t) \cos(\theta) - b \sin(t) \sin(\theta) \\y(t) &= y_c + a \cos(t) \sin(\theta) + b \sin(t) \cos(\theta),\end{aligned}$$

with $t \in [0, 2\pi]$. A set of target points is also generated on a reference circle centered at the ellipse center with an arbitrary reference radius R .

$$x'(t) = x_c + R \cos(t), \quad y'(t) = y_c + R \sin(t).$$

The homography matrix H representing the transformation mapping points on the ellipse to the reference circle is estimated using the OpenCV function `cv2.findHomography`. The homography is applied to the original image using `cv2.warpPerspective`, providing an image in which the wheel appears circular regardless of the camera angle or its position in the frame.

2.3.6 Sinkage estimation

The final step in the data processing pipeline is sinkage estimation. This is done according to the definition provided in Figure 2.11. To compute the sinkage z , the points where the predicted wheel boundary intersects the regolith surface, A and B , are required. The wheel center O and radius R are taken from the outputs of the homography transform discussed above. For now, these points must be provided by the user's manual input however the tool can be improved in future work with the addition of an automatic wheel-terrain boundary detection, as will be discussed in section 2.5. Note that these point are selected on the original image, so a perspective transformation must be applied using the same homography employed for that image.

The wheel-regolith boundary is approximated by the line AB and point C is defined as the orthogonal projection of the wheel center O onto the line AB . The sinkage can now be found as a percentage of the wheel radius: $z\% = \frac{R-|OC|}{R} \cdot 100$.

2.3.7 Performance Metrics

An important result of this research is to provide a robust ellipse fitting tool, from which the homography transform parameters are obtained. The quality of the ellipse fit should be assessed under a range of conditions, including variations of SNR (which is a tunable parameter during rover operations through the exposure time) and regolith types. The basis for comparison will be the ellipse obtained from the manual fitting tool, where the user is required to visually identify 5 points belonging to the wheel contour on the image. This result will be serve as the ground truth for comparison and assessment.

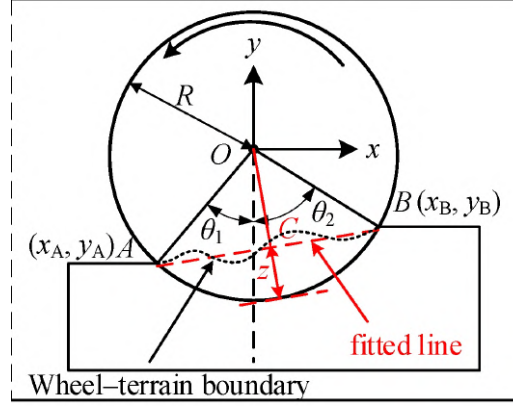


Figure 2.11: Sinkage definition [31]

To evaluate the quality of the predicted ellipse with respect to the ground-truth, two metrics are used: the Hausdorff distance and the Intersection over Union (IoU), which capture different aspects of geometric similarity.

Hausdorff distance d_H : The Hausdorff distance measures the maximum deviation between two sets of points, which in this context correspond to sampled points on the predicted ellipse and on the ground-truth ellipse. Mathematically, the Hausdorff distance between two sets A and B is defined as

$$d_H(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} \|a - b\|, \sup_{b \in B} \inf_{a \in A} \|b - a\| \right\}.$$

This metric represents the largest minimum distance between the two contours, which quantifies the worst-case geometric error between two shapes. A small Hausdorff distance indicates that every point sampled on the predicted ellipse lies close to some point on the ground-truth ellipse and vice versa. It is also important to note that the ellipses should be densely sampled to ensure the discrete distance converges to the true geometric minimum and avoid overestimating the minimum distance.

Intersection over Union IoU: While the Hausdorff distance measures the worst-case error, the Intersection over Union evaluates the area overlap between the predicted ellipse and the ground-truth ellipse. It is defined as

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|},$$

where A and B denote the interior regions of the predicted and ground-truth ellipses, respectively. IoU takes values in the range $[0, 1]$, where $\text{IoU} = 1$ indicates perfect overlap and $\text{IoU} = 0$ indicates no overlap.

This metric captures the global agreement in position, orientation, and scale. Unlike the Hausdorff distance, IoU is less sensitive to local deviations but penalises errors in the semi-major axis and overall area more strictly.

Complementarity of performance metrics: Rather than relying on a single metric, using both metrics provides a more complete and robust fitting assessment. The Hausdorff distance captures the worst-case error in the contour, while IoU measures the overall consistency of the two ellipses.

The tool user can be confident that the detected ellipse closely agrees with the ground-truth only when the indicators converge, specifically when IoU approaches 1 and Hausdorff distance approaches 0. If both or only one of the two metrics indicates a significant deviation, the fit should be considered inaccurate.

2.3.8 Model parameter tuning

A summary of the optimal parameters found through trial and error for the rear and front WheelCam algorithms is provided in Table 2.1 and Table 2.2. These values were found to perform well with the reference images, under those lighting conditions and JAXA regolith simulant. The extension of the model to unseen data, including different regolith simulants and lighting conditions, will be treated in section 2.4.

Parameter	Value
Segmentation bounds [Pa,Pd]	[0.35,1.85]
Wheel confidence area	200x200
Wheel confidence origin	[0,400]
Closing kernel	(6,6)
Opening kernel	(60,60)
Gaussian blur K, σ	(5,5), 1
Canny [minVal,maxVal]	[220,250]

Table 2.1: Rear WheelCam algorithm

Parameter	Value
CLAHE clipLimit	2.5
CLAHE tileGridSize	(10,10)
Segmentation 1 (Full image)	
Segmentation bounds [Pa,Pd]	[0.5,1.5]
Wheel confidence area	200x200
Wheel confidence origin	[400,400]
Segmentation 2 (x>700)	
Segmentation bounds [Pa,Pd]	[0.7,1.2]
Wheel confidence area	200x200
Wheel confidence origin	[200,200]
Gaussian blur K, σ	(5,5), 1
Canny [minVal,maxVal]	[220, 250]

Table 2.2: Front WheelCam algorithm

2.4 Results and Discussion

In this section, the algorithms detailed in section 2.3 will be tested at different steps of the process to assess the range of conditions for which the required performance is met and to define operational recommendations where possible. This is especially important considering the degree of uncertainty regarding the Phobos surface, including the reflectivity of the regolith.

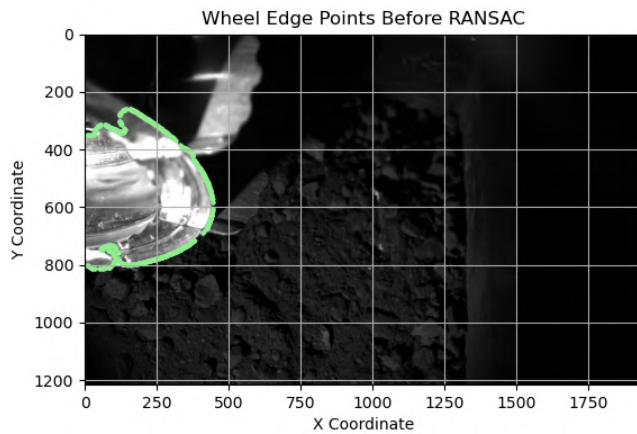
First, the rear WheelCam algorithm developed during the first semester will be applied to a set of images taken with the ISAE testbed presented in subsection 1.1.4 using different regolith simulants and different levels of wheel sinkage. Then, the front WheelCam algorithm will be

studied to determine whether a trend can be identified between ellipse fitting performance and the image's saturation across regolith types. Finally, the sinkage estimation algorithm for the front WheelCam will be compared to experimental data acquired with vertical laser in the testbed to determine the achievable sinkage estimation accuracy.

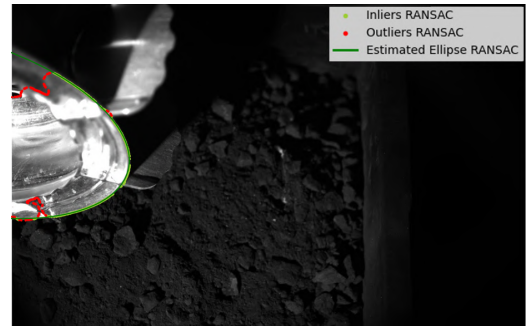
2.4.1 Analysis of rear WheelCam contour detection and ellipse fitting for different regolith simulants

The aim of this experiment is to understand whether the contours detected with different regolith simulants are sufficiently accurate for the RANSAC algorithm to produce a stable ellipse fit. In all cases, the ellipse fit was run on the detected contours with the same RANSAC parameters: `max trials = 800` and `threshold residual = 3`.

The image presented in Figure 2.12 is the baseline case for which the algorithm was developed and tuned. Due to the high contrast between the regolith and wheel, a large portion of wheel edge is successfully detected by the algorithm. This scenario is representative of the most likely conditions that IDEFIX is expected to encounter on Phobos. Despite the uncertainty regarding the regolith on Phobos, analysis of remote images taken during flybys by previous missions (including ESA's Mars Express mission in 2004) has led scientists to believe that the regolith is likely dark and fine-grained, as modeled by the JAXA regolith simulant[7].



(a) Wheel contour detected under JAXA regolith simulant conditions

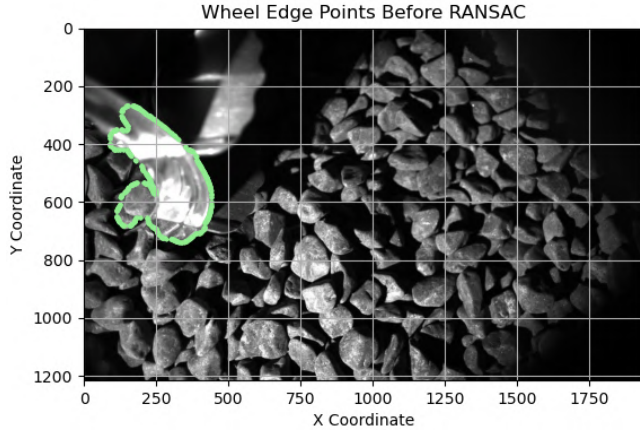


(b) RANSAC ellipse fit for the contour detected in Figure 2.12a

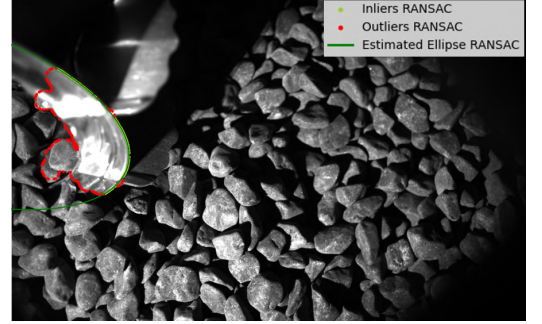
Figure 2.12: Results for the testbed rear WheelCam using JAXA regolith simulant

The image presented in Figure 2.13 presents a more challenging environment for the wheel detection process. The contrast between the wheel and gravel is less distinct due to the reflective nature of the gravel. There is also significantly more sinkage, making less of the wheel visible in the image. To obtain this result, the wheel confidence area of 200x200 pixels was kept but the origin was placed at (200,400). This shifted the confidence region away from the left edge of the image where the gravel covered the wheel.

The image presented in Figure 2.14 also poses significant challenges for the detection algorithm. Compared to the JAXA simulant, the moon regolith is much more reflective. To obtain this



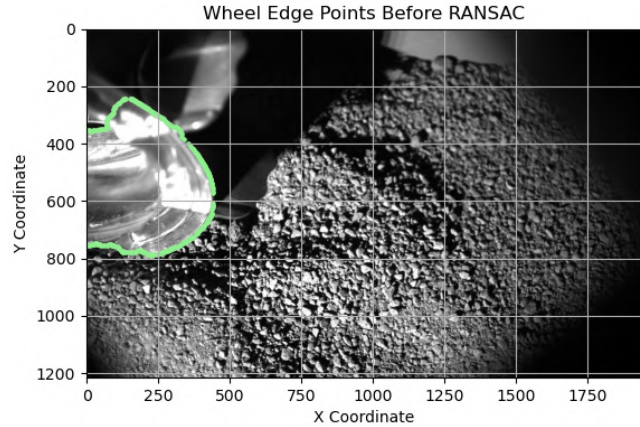
(a) Wheel contour detected under coarse gravel regolith conditions



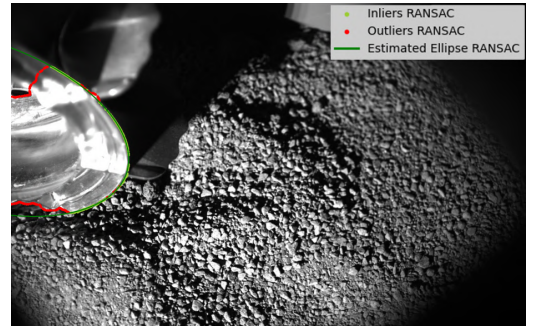
(b) RANSAC ellipse fit for the contour detected in Figure 2.13a

Figure 2.13: Results for the testbed rear WheelCam using coarse gravel regolith simulant with significant sinkage

result, the same wheel confidence area as for Figure 2.13 was chosen. Additionally, the kernel for the morphological opening transformation was adjusted to a slightly larger (65x65 elliptical) kernel in order to filter out the grousers as desired. Note that no other model parameters were altered (including the segmentation bound parameters P_a and P_d).



(a) Wheel contour detected under Moon regolith simulant conditions



(b) RANSAC ellipse fit for the contour detected in Figure 2.14a

Figure 2.14: Results for the testbed rear WheelCam using Moon regolith simulant

These three cases show that the model is able to detect the wheel contour with different regolith types that IDEFIX may encounter on Phobos. It is promising that the segmentation still identifies the wheel as the main object when encountering different levels of reflectivity of the regolith. Only small modifications of the model parameters, in particular the wheel confidence area and kernel size were made across the three cases which shows that the model is general and can be adjusted easily, which is important to ensure a smooth process when the first images arrive from Phobos.

The quality of the contours detected was sufficient for all regolith cases to fit an ellipse accurately to the wheel edge with the specified RANSAC parameters. In the next section, performance metrics will be presented to help quantify the accuracy of the ellipse fit.

2.4.2 Analysis of front WheelCam ellipse fitting under variable SNR and regolith simulants

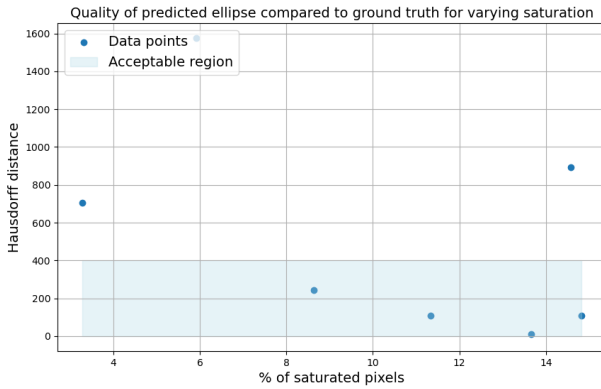
The algorithm was then applied to study the front WheelCam images with the adjustments described in subsection 2.3.3 for different regolith simulant types. The quality of the ellipses detected by the pipeline were also quantified using the metrics describes in subsection 2.3.7. This allowed to assess whether detection quality is clearly correlated to the image's saturation and, if so its effect.

JAXA simulant: Images were obtained with different levels of saturation by incrementing the brightness of the LEDs, obtaining images ranging from 2% to 15% saturation - i.e. the fraction of saturated pixels. Note that the images were taken with relatively low sinkage, and results might differ for varying levels of sinkage. For each image, the segmentation model and RANSAC fitting algorithm was run with the same parameters (number of maximum iterations and threshold residual), obtaining a prediction of the ellipse. For each image, the ground-truth ellipse was obtained with the manual tool and both were compared using the above mentioned performance metrics.

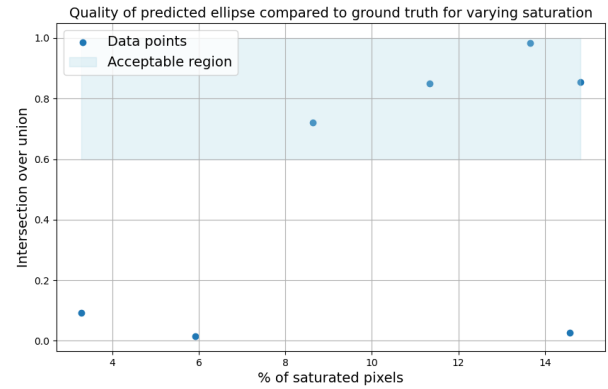
In Figure 2.15a and Figure 2.15b, the Hausdorff distance and IoU between the predicted ellipse and ground-truth are presented as a function of the level of saturation in the raw image.

An acceptable region was taken for a Hausdorff distance below 400 and over 0.6 for the IoU; these are represented by the light-blue shaded area in the graph. Considering this, the optimal range of saturation level is located between 8% and 15%, with the closest fit provided by the image at **13.6%** (Hausdorff = 11.1, IoU = 0.98).

The obtained ellipse fit for this saturation level is provided in Figure 2.16 to help visualise the quality of the fit.



(a) Hausdorff distance vs. saturation



(b) Intersection over union vs. saturation

Figure 2.15: Ellipse fit performance metrics as a function of saturation, JAXA regolith simulant

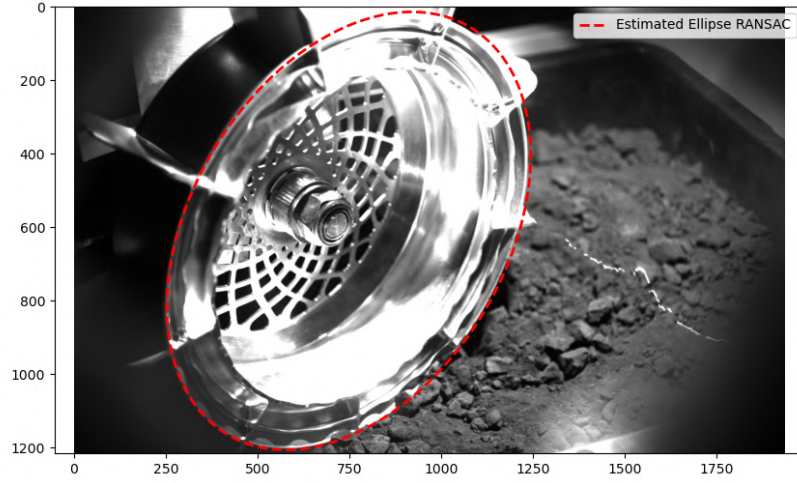
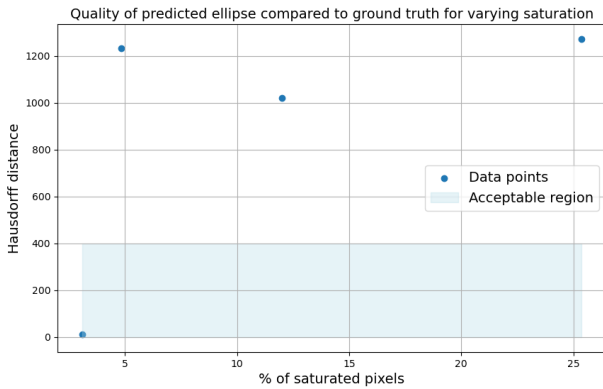
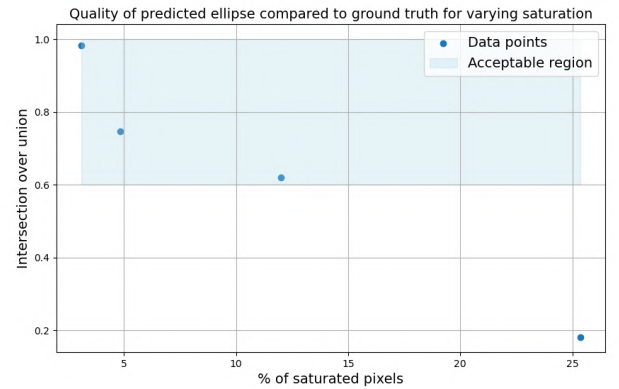


Figure 2.16: RANSAC ellipse fit with JAXA regolith at 13.6% saturation

Lunar regolith simulant: The same experiment was performed of the Moon regolith simulant, obtaining images ranging from 3% to 26% saturation. The ellipse fit quality at each point is presented in Figure 2.17a and Figure 2.17b. In this case, it is clear that only the first point at 3% (Hausdorff = 10.7, IoU = 0.98) provides an appropriate ellipse fit as the Hausdorff distance for the other points is very large. The IoU metric however shows an interesting trend of worsening fit quality with increasing saturation past this point, following a linear trend.



(a) Hausdorff distance vs. saturation analysis



(b) Intersection over union vs. saturation analysis

Figure 2.17: Ellipse fit performance metrics as a function of saturation, Moon regolith simulant

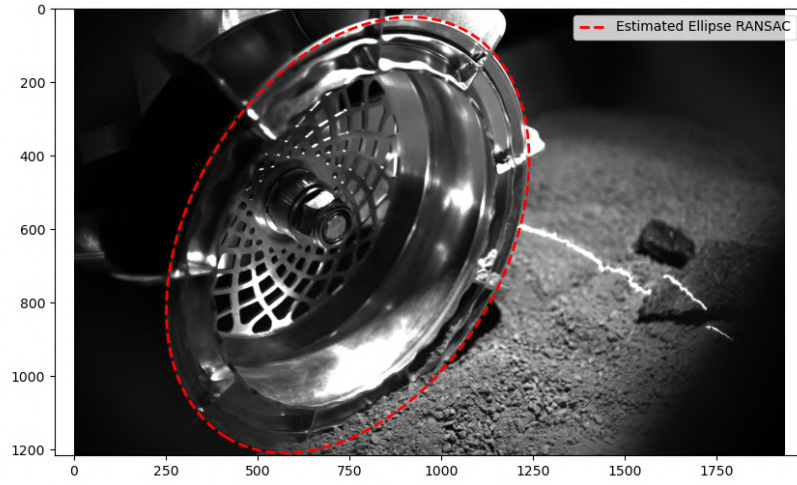


Figure 2.18: RANSAC ellipse fit with Moon regolith at 3.6% saturation

These two experiments show that the model provides better results overall with the JAXA regolith simulant, displaying a better robustness on this dataset, for which an appropriate ellipse is obtained for multiple points around the ideal saturation.

We can also conclude from this analysis that the model prefers a different ideal saturation for different regolith types, as summarised in Table 2.3 for the experiments carried out. A more reflective regolith (ie. Moon regolith simulant) requires lower saturation levels, and vice versa. This conclusion has important operational consequences: the ideal saturation level is a crucial parameter which depends on the reflectiveness of the regolith and thus has to be tuned with initial in-situ image acquisitions to provide a reliable and robust segmentation model for contour detection and ellipse fitting.

Regolith Simulant	Optimal saturation range	Best fit saturation
JAXA	8% - 15%	13.6%
Moon	-	3%

Table 2.3: Optimal saturation level for various regolith simulants

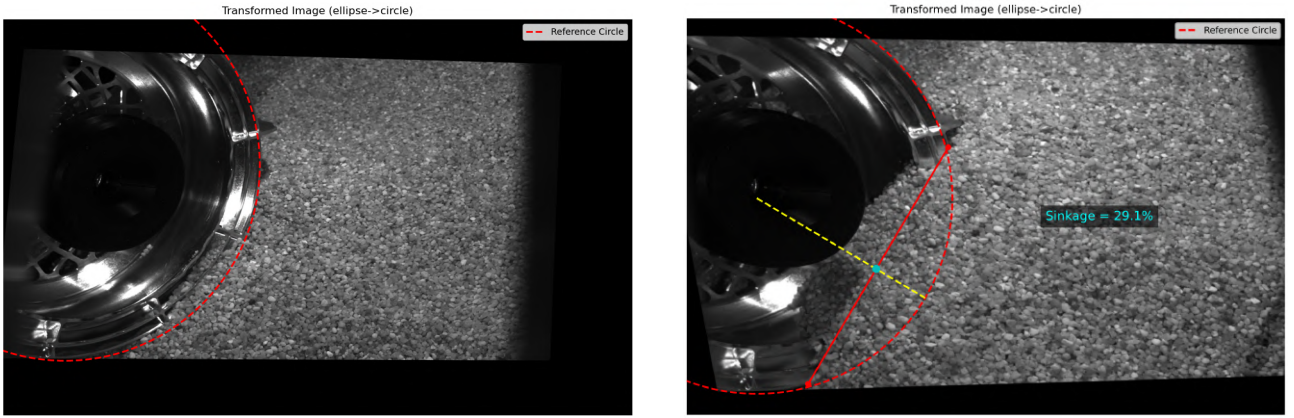
2.4.3 Validation of sinkage estimation using testbed laser measurements

The purpose of this section is to quantify the accuracy of the sinkage estimation provided by the image processing pipeline, and relate the accuracy back to the ellipse fitting precision. The sinkage estimation will be compared with experimental data from the vertical laser in the testbed, which measures the vertical position of the wheel in the testbed from a fixed datum.

First an image and laser measurement were acquired at the zero-sinkage position of the wheel, with the outer wheel rim resting on top of the regolith. Then, the wheel was sunk in the regolith

(making sure not to disturb the regolith in the region around the wheel) and a second image and measurement were acquired. The difference in y-position of the wheel between the two acquisitions corresponds to the wheel sinkage, assuming the regolith interface line is perfectly horizontal.

The image processing pipeline was run on the image with the sunk wheel, including contour detection, ellipse fitting, homography transform and sinkage estimation. Both image acquisitions are shown with their homography transform applied in Figure 2.19a and Figure 2.19b respectively.



(a) Image acquisition at zero sinkage

(b) Sinkage estimation from automatic pipeline

Figure 2.19: Image acquisition positions used for the sinkage validation testset

The laser measurements were: Measurement 1 (no sinkage) $y = 90.32$ mm and Measurement 2 (with sinkage) $y = 119.67$ mm. From these measurements, the sinkage was computed as 29.35 mm corresponding to 33.74% of the wheel, using $\frac{y_{sunk} - y_0}{R_{wheel}}$ with $R_{wheel} = 87$ mm.

On the other hand, the result obtained for this image from the pipeline was 25.37 mm (or 29.1% of the wheel). This corresponds to an absolute error of 3.86 mm and a relative error to the true sinkage (as measured in the testbed) of $\mathbf{E_{rel} = 13.75\%}$.

It is interesting to relate the final sinkage estimation to the error in the predicted ellipse compared to the manual ground truth as studied before. Applying the sinkage estimation pipeline on the manual ellipse fit yields a sinkage of 22.79 mm, or 26.2%. This allows to deduce that the ellipse fitting algorithm is associated to an error bar of $\mathbf{E_{abs} \pm 2.52}$ mm. These differences and the fitted ellipse's quality are summarised in Table 2.4.

Method	Sinkage	Absolute error	Relative error %	d_H	IoU %
Manual ellipse fit	22.79 mm 26.20 % R_{wheel}	-	-	-	-
Automatic ellipse fit	25.37 mm 29.10 % R_{wheel}	2.52 mm	11.07	257	74

Table 2.4: Correlation between ellipse fit error and sinkage estimation, assuming manual fit as reference

The agreement between the two ellipses is surprisingly low given how well the contour is being tracked on the portion of the wheel that is visible in Figure 2.19 and highlights a key issue: if only a fraction of the wheel is visible in the image, then the ellipse fit on the remaining section of the wheel is unreliable, in particular with regards to the semi-major axis.

To illustrate this point, the manual and predicted ellipses are shown in Figure 2.20. It can be seen that in both cases, the wheel contour is tracked on the visible portion, but there is insufficient data to accurately predict the wheel contour where it is not visible.

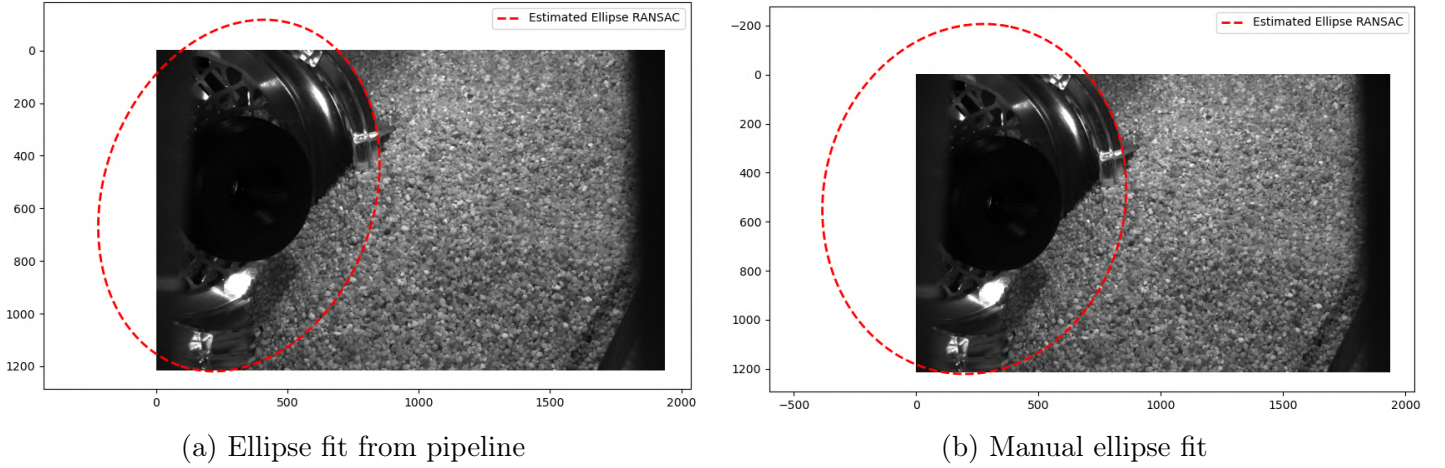


Figure 2.20: Comparison of manual and automatic ellipse models

The camera view used for the sinkage validation test is the one from the new setup following the changes discussed in subsection 1.1.4, which has the disadvantage of having a smaller portion of the wheel visible, although it is more realistic of the true camera angle on the IDEFIX rover.

Finally, when comparing the sinkage estimation provided by the automated pipeline against that given by the laser measurements gives an error bar of $\mathbf{E}_{\text{abs}} \pm 4.04$ mm. This difference places the sinkage given by the laser measurement beyond the error bar associated with the uncertainty resulting from the ellipse fitting process. This suggests that additional errors arise from elsewhere than the image processing algorithm, possibly from the experimental protocol and the assumption that the laser profilometer provides true sinkage. Indeed, the experiment carried out is not perfect as the process of sinking the wheel leads to a disturbance of the regolith region surrounding the wheel. Compressing the simulant under the wheel leads a partial collapse of the regolith in front of the wheel, and thus an underestimation of sinkage by the pipeline is expected compared to the laser measurement.

Method	Sinkage	Absolute error	Relative error %
Laser measurement	29.35 mm	-	-
	33.74 % R_{wheel}		
Automatic ellipse fit	25.37 mm	4.04 mm	13.75
	29.10 % R_{wheel}		

Table 2.5: Summary of sinkage validation test results, using laser measurements as ground truth

2.5 Image Processing Pipeline - Future perspectives

The main area that needs work in the future is to improve the reliability of ellipse fitting under the revised camera angle in the testbed, which is more representative of the true images IDEFIX will capture. This camera angle proves to be more difficult to work with as a significant portion of the wheel edge is not captured, making ellipse fitting unreliable even when the contour is well detected. A possible solution would be to impose lower and upper bounds on the predicted semi-major axis based on an educated guess from the camera position, viewing angle and its approximate distance to the wheel.

The next step would be to improve the pipeline, accounting for the changes due to the new camera angle, and to correct the sinkage offset that is not observable leveraging additional methods. In particular, the validation dataset should be expanded with data from the new testbed setup to determine whether the relation observed between ellipse fit quality and sinkage estimation accuracy follows a trend under varying sinkage depth. More formal work should also be dedicated to improving the experimental protocols ensuring that the sinkage obtained using the vertical laser can be used as ground truth without major disturbance of the surrounding region used for this vision-based method.

Finally, the last aspect of this research that can be improved is to fully automate the pipeline. Currently, the sinkage estimation tool relies on user inputs to define the interface between the wheel and regolith by selecting two points. A possible way to automate this step would be to combine it with a segmentation model that can distinguish wheel surface from regolith surface. Overlaying the detected ellipse with such a segmentation mask would allow defining these points without manual inputs. Figure 2.21 shows an example of what the output of this segmentation model could be like based on work being carried out currently within the SSPA team at ISAE SUPAERO.



Figure 2.21: Wheel and regolith binary segmentation (D. Billot)

3 Material Composition Detection

3.1 Introduction

This experiment aims to validate whether material composition can be detected using the IDEFIX rover's panchromatic camera under controlled LED illumination. This method relies on the use of the IDEFIX rover's panchromatic WheelCams and three LEDs of wavelengths 590 nm, 720 nm, 880 nm, to illuminate the surface and measure the relative brightness of the reflected light.

Given that the surface material on Phobos is expected to be composed of multiple elements and varying in grain sizes as explained in subsection 1.1.2, the spectral response will be affected not only by the composition, but also by surface texture. This complexity was tackled in Semester 2, as will be explained in subsection 3.5.1, with a simplified model that employed coloured papers as samples. After demonstrating the possibility of obtaining valuable information from the papers, the investigation has continued towards geological materials.

By comparing the brightness values obtained from the images with known spectral curves for different materials, we aim to recognise the composition of the samples. This model will further explore the feasibility of using cameras and LEDs for scientific purposes. Considering the complexity of the problem, the results obtained in this report will serve as a basis for future investigations and the development of more complex models.

3.2 Objectives

Although the semester objectives have already been generally defined in section 1.3, let us specify exactly what the aims are:

- Perform the experimental tests on geological materials using the model concept of Semester 2 and evaluate the results. Critically state the assumptions made and limitations of the approach.
- Perform the forward approach: determine whether the relative brightness derived from the WheelCams and LEDs is coherent with reference reflectance spectroscopy data.
- Evaluate the feasibility of the experimental method and state the modifications needed for the success of future testing.

The objectives remain similar to those of the previous semester, as the a very similar experimental procedure was employed. However, although the feasibility of the methodology was established using coloured paper samples, this does not guarantee its applicability to geological materials, which present significantly greater complexity. Therefore, the same objectives were retained to reassess the feasibility of the approach under these more challenging conditions. All objectives have been completed and will be discussed in detail in section 3.5.

3.3 Theoretical Background

3.3.1 Key Concepts

All of the definitions in the following paragraphs will be the definitions of the concepts in the specific application of this research project. Some concepts may have different definitions whether referring to radiometry, photometry, astronomy, among others.

This project focuses mainly on brightness and reflectance. Brightness will refer to the amount of "whiteness" on our greyscale images. It will be a value from 0 to 255, 0 indicating complete blackness and 255 indicating complete whiteness. This value will be measured directly from our image during the data processing. Reflectance, on the other hand, is a proportion of light that a surface reflects relative to the total light received. It is broadly used in spectral studies and it is key in the interpretation of the brightness values that will be recorded on this project. Differently from last semester, reflectance will be computed thanks to the use of a calibrating panel. It is important to define absorption as well, which refers to the proportion of light taken in by the material, rather than reflected. By identifying patterns in reflectance and absorption, material properties and composition can be identified and associated with said material. During this semester, the investigation into the feasibility of identifying these properties through the brightness from the WheelCam images is continued.

Another key concept will be sensitivity. Panchromatic cameras, as explained in subsection 1.1.3, receive information from various wavelengths, however this information is not always received equally. The sensor has a spectral sensitivity curve as illustrated in Figure 3.1, meaning it responds more to certain parts of the spectrum and less to others. This affects how the final image appears since brighter or darker areas may not only depend on how much light is present, but also on which wavelengths dominate the scene.

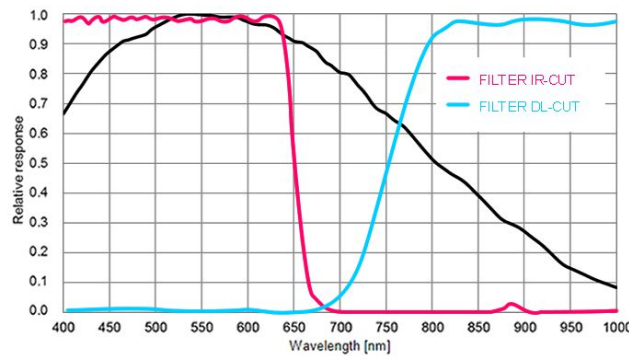


Figure 3.1: WheelCam's detector sensitivity in terms of wavelength [32]

In Figure 3.1, the black curve represents the intrinsic spectral sensitivity of the camera sensor, which spans both the visible and the near infrared wavelength ranges. The red curve, represents a filter that suppresses the infrared radiation while allowing visible light. The blue curve, on the other hand, represents a filter that suppresses visible wavelengths and transmits near infrared light.

3.3.2 Integration of key concepts

Recapitulating the theory explained in section 1.1, as the WheelCams are panchromatic, as well as the testbed WheelCams, there is no way to distinguish in which wavelength the data is received.

Combining the environment by illuminating it with a LED in a narrow spectral range will mean that the image received will consist of the response of the environment to said wavelength. In other words, a visual representation of reflectance and absorption at a concrete wavelength will be obtained.

Although this image provides a visual representation of the spectral properties, the actual values of reflectance and absorption cannot be obtained directly through this process. Only brightness values can be extracted from the greyscale image. However, the use of a calibration panel makes it possible to compute reflectance values, advancing the investigation conducted last year, in which conclusions could only be drawn from brightness measurements.

$$f(x, y) = i(x, y)r(x, y) \quad (3.1)$$

Equation 3.1 is obtained from [33], where it is explained that the brightness of an image $f(x, y)$ is related proportionally to reflectance $r(x, y)$ by illumination components $i(x, y)$, where x and y represent the position on the image (pixels). The latter is determined by the illumination source, whilst the reflectance is a property of the material under illumination. Moreover, as seen in subsection 3.3.1, the sensitivity of the camera's detector must be considered. Therefore, Equation 3.1 becomes Equation 3.3.

$$B(\lambda) = L(\lambda) \times R(\lambda) \times S(\lambda) \quad (3.2)$$

In this equation, the relationship between brightness and reflectance can be observed in terms of wavelength (λ), where $B(\lambda)$ depicts brightness; $R(\lambda)$, the reflectance; $L(\lambda)$, the illumination source intensity and $S(\lambda)$, the camera's sensitivity.

The sensitivity is a factor that depends on the camera as presented in subsection 3.3.1, and has a value between 0 and 1. As an approximation of Figure 3.1, the values of sensitivity factors for the LED's wavelengths are given in Table 3.1.

Wavelength	Sensitivity
590 nm	0.98
720 nm	0.75
880 nm	0.3

Table 3.1: Panchromatic's camera sensitivity for a given wavelength

However, the investigation conducted during Semester 2 suggests that the sensitivity correction is already included in the camera's internal processing. If this correction had not been applied, the brightness values recorded by the camera would have been significantly affected

by the strong variation in sensitivity across wavelengths, particularly in the near-infrared region. In that case, the trends measured by the camera would not match those measured by the spectrometer.

Since the brightness trends obtained from the camera closely matched the spectrometer measurements, it was concluded that the sensitivity factor is already accounted for in the camera's correction parameters, and therefore does not need to be applied separately during the data processing. However, the sensitivity for infrared is significantly low, which should be taken into account on the interpretation of the measurements.

Another parameter that must be taken into account is the integration time of the camera. The detector of the panchromatic camera does not measure the instantaneous light intensity, but rather accumulates the photons received during a given exposure period. This exposure period, $t_{integration}$, directly affects the measured brightness values: the longer the integration time, the more light is collected by the sensor and the higher the resulting brightness value. This allows the values to be normalized while ensuring that the sample remains visible in images illuminated by different LEDs.

Therefore, the brightness recorded by the camera is proportional not only to the illumination intensity, the reflectance of the material, and the spectral sensitivity of the detector, but also to the integration time. Equation 3.3 can be therefore rewritten as:

$$B(\lambda) = L(\lambda) \times R(\lambda) \times S(\lambda) \times t_{integration} \quad (3.3)$$

The illumination source intensity of each LED is the unknown quantity of interest, in red in Equation 3.3. Thanks to the use of a calibrating panel, it is possible to estimate this value. The calibration panel has a nearly constant reflectance of 0.97, which means that almost all the light it receives is reflected. By measuring the brightness of the calibration panel in the images, the intensity of each LED can be determined. Therefore, Equation 3.3 can be rewritten as:

$$B_{cal}(\lambda) = L_{LED}(\lambda) \times R_{cal}(\lambda) \times t_{cal} \quad (3.4)$$

where B_{cal} is the brightness measured in the image of the calibration panel, R_{cal} is the known reflectance of the calibration panel (0.97), and L_{LED} represents the illumination intensity of the LED at wavelength λ .

From this relation, the LED illumination intensity can be estimated. Once L_{LED} is determined, it can be used to compute the reflectance of the sample using the same formulation as before. In this case, however, the brightness is obtained directly from the image of the sample, and the corresponding integration time is taken from that image. The reflectance is then calculated according to 3.6.

$$B_{sample}(\lambda) = L_{LED}(\lambda) \times R_{sample}(\lambda) \times t_{sample} \quad (3.5)$$

$$R_{sample} = \frac{B_{sample}}{t_{sample}} \times \frac{1}{L_{LED}} = \frac{B_{sample}}{t_{sample}} \times \frac{t_{cal}}{B_{cal}} \times R_{cal} \quad (3.6)$$

where B_{sample} is the measured brightness for the sample.

Through the previous discussion it has been shown that, in theory, brightness holds a proportional relationship to reflectance under controlled conditions. This allows brightness values obtained from the images to be used as direct indicators of material reflectance. In order to draw significant conclusions, these values must be compared to experimental spectral studies of the desired material for testing.

For each available LED, an average brightness value will be obtained at small region of pixels. By computing reflectance with Equation 3.6, analysing the trend of these values across wavelengths and comparing them with known reflectance curves from spectrometric studies, the recognition or differentiation of different materials on the image is possible. With the collaboration of a researcher from the Centre de Recherches Pétrographiques et Géochimiques (CRPG), the samples were provided as well as the reference curves that were used as the comparative data for our experiments, taken with multispectral cameras at the research laboratory. As additional comparative data, United States Geological Survey (USGS) [34] spectral libraries will be used for materials of special interest.

While the exact composition of Phobos' surface materials is unknown, variations in brightness at specific wavelengths can still provide relevant information regarding space weathering and material composition as indicated in subsection 1.1.2. More specifically, in the wavelength range at which the cameras and LEDs operate, the images obtained can be use to reveal the presence of minerals such as olivine and pyroxene that can be detected in the 800 to 1000 nm region [7].

3.4 Methodology

Due to the complexity of the problem, it was necessary to begin testing with a simplified model to validate whether material differentiation is feasible using spectral methods. During previous work, coloured papers were tested in a straightforward approach. The limits of this preliminary method were reached, and the experiments this semester were extended to geological material samples. Although this represents a step towards more realistic conditions, the work remains far from testing actual regolith samples, as it only introduces limited surface complexity. In this case, the sample presents a non-uniform surface, but it does not capture the full range of properties of regolith, such as its granular structure, variable particle size distribution, porosity, and complex light scattering behaviour.

3.4.1 Experimental Test Set-up

The following equipment has been used to perform the experimental tests.

- ISAE-SUPAERO testbed (as described in subsection 1.1.4)
 - Four monochromatic LEDs: Only one of the White (used as reference), Yellow (590 nm), Red (720 nm) and IR (880 nm).

– Panchromatic testbed Rear WheelCam

- Meteorites (specified in Appendix A)
- Analogue Samples (specified in subsection 3.5.3)
- Calibration panel

Only the rear WheelCam was used, as it provides a larger field of view of the sample. The front camera captures more of the wheel itself, whereas the rear camera captures more of the regolith, allowing greater flexibility in positioning the sample during testing.

The setup consisted of placing each geological material sample in the test area, on the location that simulated the regolith surface. In contrast to the previous semester, the experiments were conducted without the use of a spectrometer.

An important change introduced during this semester's work was the addition of a calibration panel. This panel was implemented after it was observed that the LEDs exhibited different light intensities. By using the calibration panel, this variation can now be accounted for and the reflectance computed. As a result, the measurements obtained this semester could be directly compared with known spectral curves, as discussed later in subsection 3.3.2. The set-up can be observed on Figure 3.2.

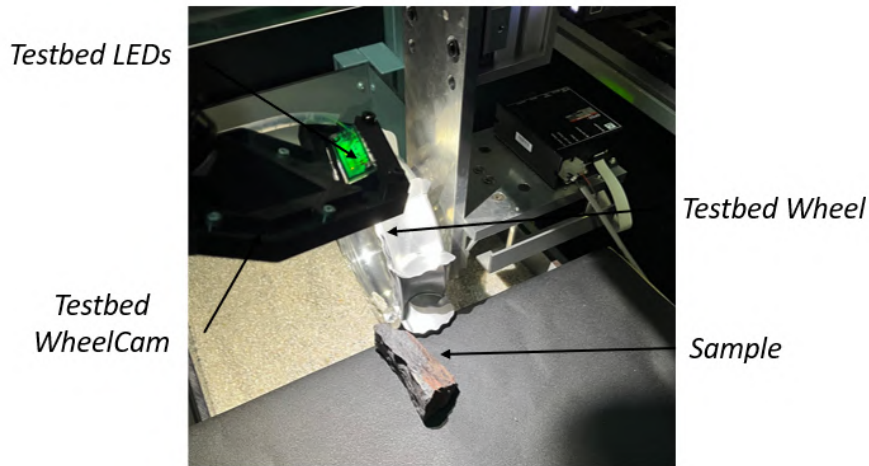


Figure 3.2: Experimental Set-Up example, with white LED on and analogue material sample

During the initial stages of the investigation, a misalignment of the LEDs became apparent, as well as differences between the testbed equipment, such as the camera optics and the positioning, and the actual conditions on the IDEFIX rover. As a result, the testbed configuration was updated during the course of the study.

3.4.2 Experimental Test Protocol

Once the sample was positioned so that it was illuminated by all three LEDs, the testbed blackout panels were added to eliminate interference from ambient light. However, due to minor gaps in the enclosure, some ambient light contamination may still have occurred. This should therefore be acknowledged as a potential source of measurement error and a limitation of the experimental setup.

For each sample, four measurements were taken using the panchromatic camera, one under each LED (white, 590 nm, 720 nm, and 880 nm). In addition, measurements were performed for each LED using the calibration panel. During the analysis, it became evident that measurements of the calibration panel at different altitudes were also required, and these were subsequently acquired.

Following the testbed update described in subsection 3.4.1, additional measurements were carried out using the revised configuration. These were acquired following the same procedure outlined above to ensure consistency with the previous dataset.

3.4.3 Data Processing and Analysis

For each LED, a brightness value was extracted from the greyscale image by averaging the intensity over a 7×7 pixel area centred on the selected point of interest for each sample. These points were chosen either based on distinctive features that could provide insight into the sample's origin, or on clearly identifiable characteristics to enable comparison with the CRPG data. The use of an area average reduces the impact of random noise and ensures consistency across measurements. With the addition of the calibration panel, the true reflectance of the material can then be derived from the measured brightness, as explained in subsection 3.3.1.

In these experiments, two main families of samples were used: meteorites and analogue samples. For the meteorite samples, spectral cubes provided by CRPG were used. For the analogue samples, spectral libraries were used instead. These included spectra from CRPG, obtained from the same rock samples, as well as reference spectra from the U.S. Geological Survey [34], which represent typical spectral signatures of specific material compositions.

To understand the data processing methodology, it is necessary to introduce the software ENVI, as well as the concepts of spectral libraries and spectral cubes.

ENVI is a software platform used to analyse images that contain spectral information. It allows users to process images, study how materials reflect light at different wavelengths, and compare these measurements with reference data from spectral libraries in order to help identify materials. A spectral library is a collection of reference spectral signatures of known materials. Each entry of the library represents how a specific material reflects or emits light across different wavelengths. If the entry comes from a real sample, it represents the average reflectance of a chosen area in the material. Finally, a spectral cube is a three-dimensional dataset that contains both image and spectral information. For every pixel in the image, the dataset stores how the material reflects light at different wavelengths. Because this information is recorded for each pixel, spectral cubes are very large data files. However, it is possible to analyse individual pixels

or calculate average values over selected areas of the image. Both of these references will allow us to compare our measurements and classify them. In the following images, a representation of a spectral library and a spectral cube on ENVI can be observed.

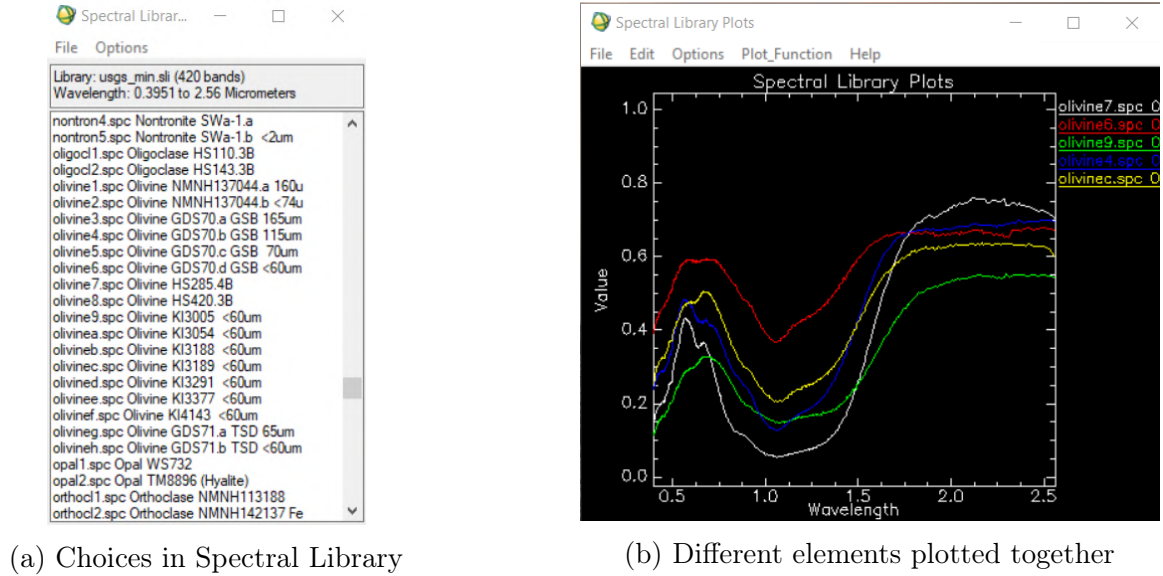


Figure 3.3: Example of USGS Spectral Library [34] on ENVI Classic

The spectral cube is visible on the left of Figure 3.4, with two additional zoom windows on the bottom centre and bottom right row. On the top centre row, the spectral profile of an average of 7 pixels around the one selected on the smallest zoom window is depicted. On the top right row, the list of all the bans available.

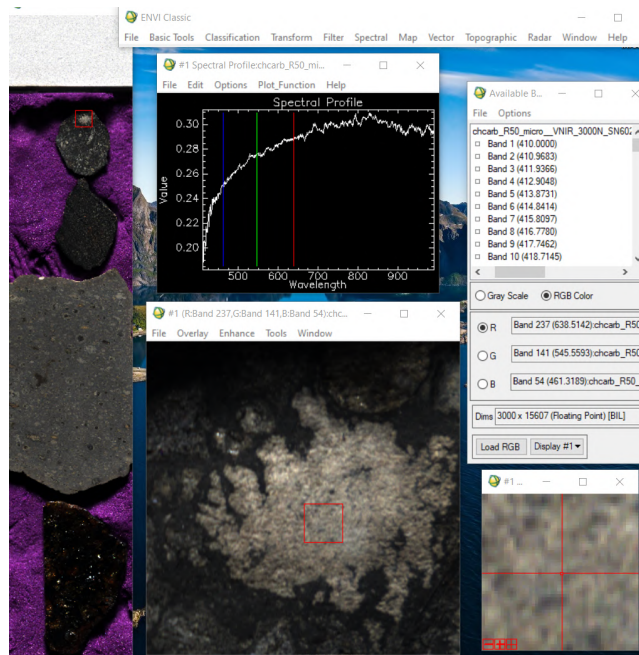


Figure 3.4: Spectral Cube depiction with spectral profile of an average window (7 Pixels)

The use of these references provided a ground truth against which the experimental results could be compared, allowing the feasibility of using the combination of LEDs and a panchromatic camera to distinguish material compositions to be evaluated.

To analyse the similarity in trends between the experimental measurements and the reference data, several approaches were applied. First, to facilitate visual comparison, both datasets were normalized and plotted together in order to observe similarities between the curves. Then, quantitative comparison was performed using the Pearson Correlation coefficient and Cosine Similarity coefficient.

The Pearson correlation coefficient [35] measures the linear relationship between two sets of variables, however, it does not capture nonlinear or monotonic relationships. It is represented by the following formula:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (3.7)$$

where x_i and y_i are the individual values from the two datasets being compared (brightness and intensity), and $\bar{x}$ and $\bar{y}$ are their respective means. The result r ranges from -1 to 1 , where:

$r = 1$ indicates perfect positive correlation

$r = -1$ indicates perfect negative correlation

$r = 0$ indicates no linear correlation

The Cosine similarity [36] measures the angle between two vectors, ignoring the magnitude. It ranges between 0 and 1 , where the closeness to 1 indicates a more similar orientation. It is useful for comparing shape and pattern rather than scale however, it disregards magnitude which might influence our ability to distinguish between materials.

$$\text{CosSim} = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}} \quad (3.8)$$

where x_i and y_i are the individual values from the two datasets being compared (brightness and intensity).

The Pearson correlation coefficient is significantly useful in this case because it measures how two variables change in a linear and proportional manner. This is exactly what was expected of the results, to have a proportionality relation between light intensity and brightness. On the other hand, cosine similarity captures the angle between the vectors of values from brightness and spectrometer, without looking at the magnitude. It measures whether the variation of this angle is similar. This makes the combination of both variables complementary and useful.

3.5 Results and discussion

3.5.1 Experimental results on Coloured Papers: Preliminary model

The first semester of this project focused on the same scientific objectives that continue to guide the current work, but through a simplified experimental approach. Initial experiments were conducted using coloured paper samples to provide a preliminary validation of the methodology and to evaluate whether the brightness measurements under different LED wavelengths could reproduce the spectral trends obtained from a spectrometer.

Spectrometer measurements under white LED illumination were first obtained to characterize the spectral trends of the papers, seen in Figure 3.5.

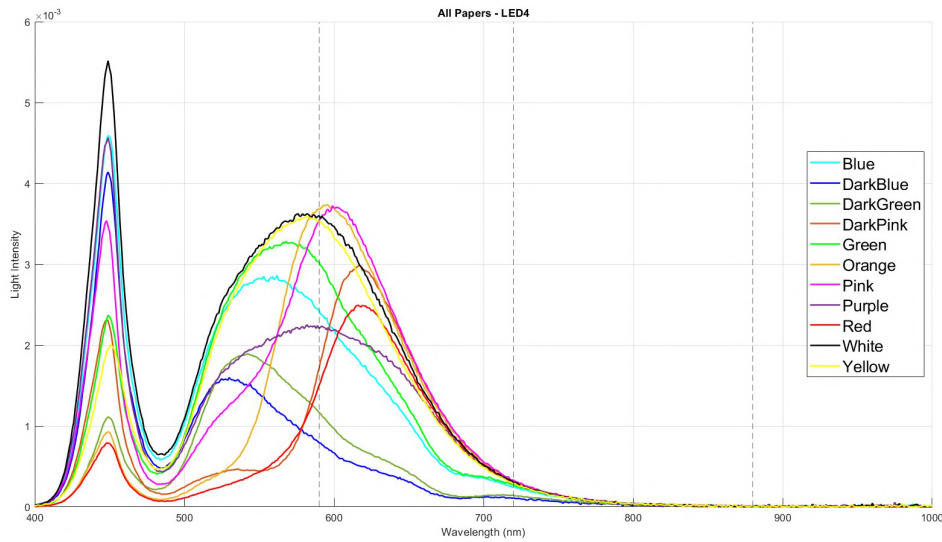


Figure 3.5: Spectral curves of the coloured papers under white LED illumination.

These measurements provided insight into the spectral behaviour of the different coloured papers and helped identify which samples would yield the most informative results. For instance, dark green and dark blue papers exhibited very low reflectance in the infrared range, making them less distinguishable in those wavelengths and therefore less useful for analysis compared to the others.

In the next test conducted, brightness values recorded by the camera under the three LEDs (590 nm, 720 nm and 880 nm) were then compared with the spectrometer measurements in order to evaluate whether the camera could reproduce the general spectral trends. These measurements were conducted prior to the introduction of the calibration panel. However, since both the spectrometer and the images were acquired under the same illumination conditions, the LED intensities were consistent across measurements, which allowed for a meaningful comparison of the results.

The results showed that the brightness measurements were generally able to reproduce the overall spectral trends of the papers. However, attempts to identify the papers based only on

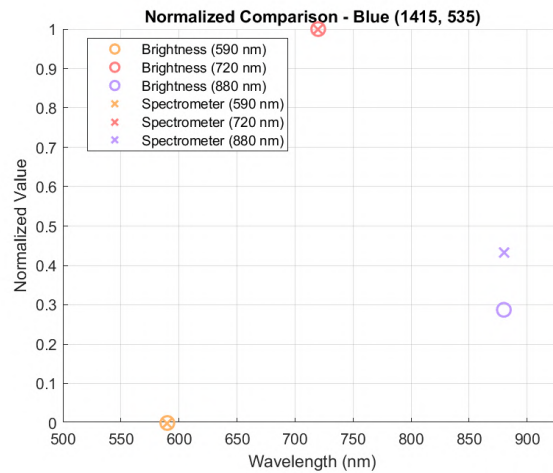


Figure 3.6: Example comparison between normalized brightness and spectrometer measurements.

their brightness measurements were not reliable.

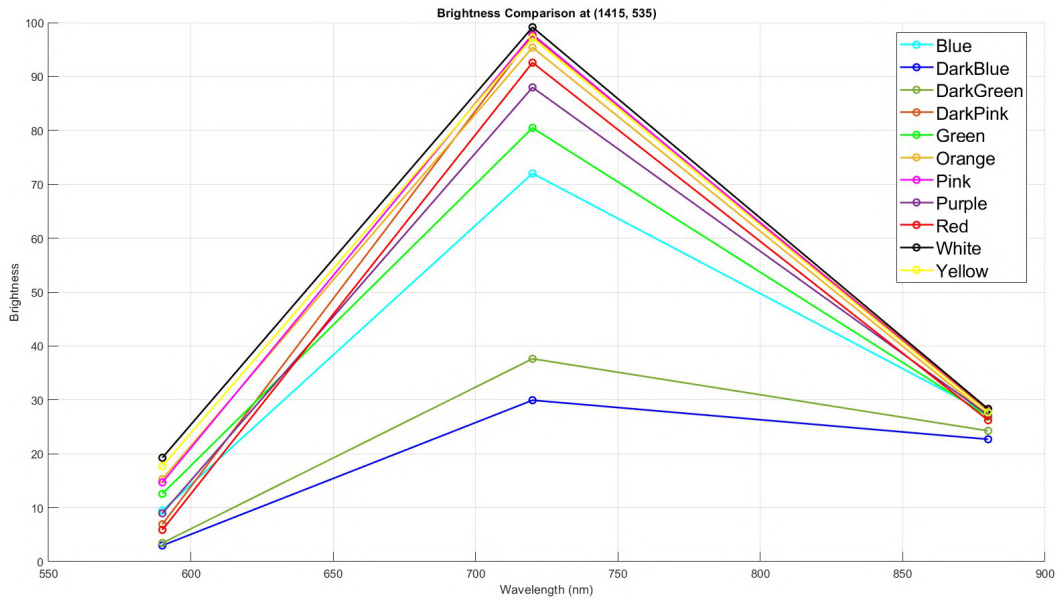


Figure 3.7: Brightness trends for all papers.

This limitation was assumed to be due to the low spectral resolution of the system, which relies on only three wavelength measurements, and to the similarity of the spectral trends observed across most papers. In addition, variations in LED intensity and the absence of a calibration panel prevented the computation of reflectance values, making the comparison dependent on relative brightness measurements.

These observations highlighted the need for improvements in the experimental setup. In particular, the introduction of a calibration panel and the transition to solid material samples were

identified as key steps for advancing the investigation during the second semester.

3.5.2 Experimental Results on Geological Samples: Meteorites

The main goal of this project was to investigate whether the brightness values recorded by a panchromatic camera under three LEDs could be used to replicate or approximate the reflectance trends of the samples. During this semester, the calculation of reflectance became possible through the use of a calibration panel. Nevertheless, the main objective remains to compare the trends of the measured data and the reference data to determine whether they show meaningful similarities.

To investigate the optical characteristics of each meteorite slice, images were acquired following the protocol described in subsection 3.4.2. The meteorite samples were available only for a limited period of time, and therefore only a single set of measurements could be obtained.

Initially, the plan was to compare these measurements either with spectrometer measurements, as performed during the first semester, or with data acquired simultaneously using a multispectral camera from CNES under the same experimental conditions. However, the spectral cubes were made available at a later stage, while the processed data from the multispectral camera was never transferred to us. Furthermore, spectral cube data was not available for all the meteorite slices, and therefore only the samples for which comparative measurements existed could be included in the analysis. In addition, for some meteorite slices the face that was imaged differed from the face represented in the spectral cube measurements, which made a direct comparison impossible. These samples were consequently excluded from the study.

Although these constraints limited the available dataset, they reflect the practical challenges often encountered in experimental investigations, particularly when rare opportunities such as working with meteorite samples arise and time does not allow for extensive experimental planning or data processing preparation.

Finally, from the initial set of 12 meteorites, named according to the order in which the experiment was conducted, only the following samples were included in the processing: M3, M5, M6, M8, M9, M10, M11, and M12. Figure 3.8 and Figure 3.9 are examples of the comparative study made. The corresponding images of all analysed meteorites are provided in Appendix A.

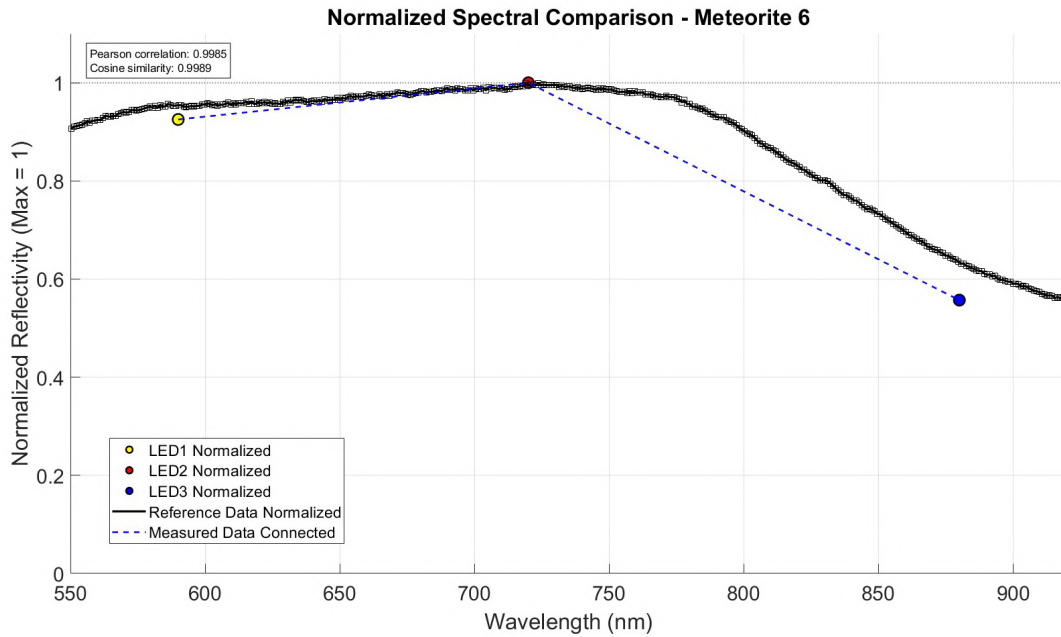


Figure 3.8: Example M6: Normalized comparison of calculated measurements and spectral cube measurements.

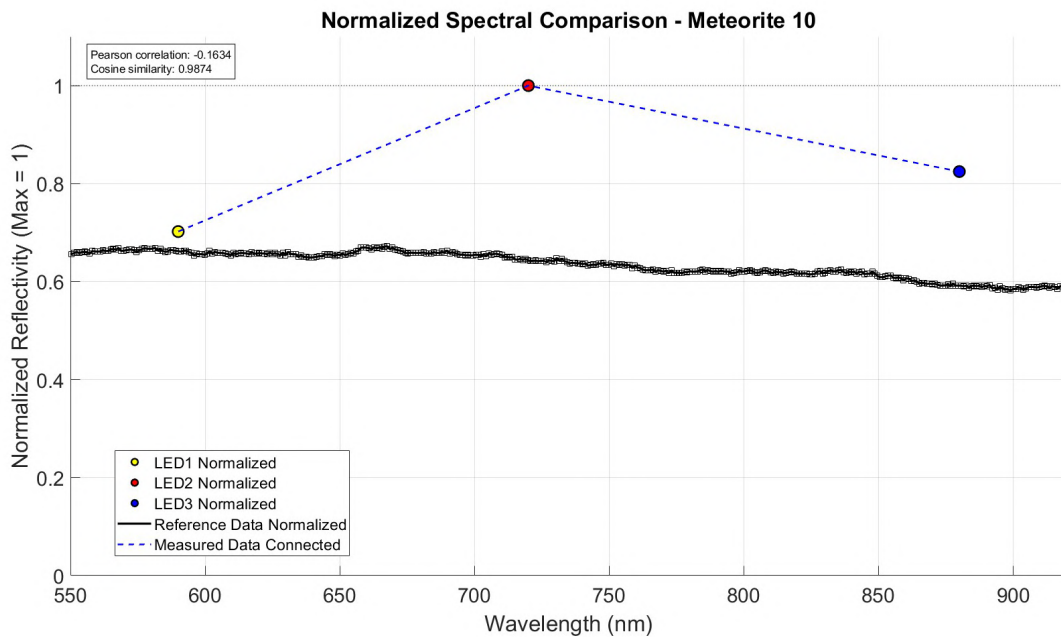


Figure 3.9: Example M10: Normalized comparison of calculated measurements and spectral cube measurements.

The results obtained for all the meteorites analysed are summarized in Table 3.2. These metrics were computed using an average window of 7×7 pixels for the multispectral images and 15×15 pixels for the spectral cube data. The influence of the averaging window size on the

measurements has also been investigated and will be discussed later on in the report.

Meteorite	Pearson	Cosine
M3	0.8799	0.9975
M5	0.8066	0.9969
M6	0.9987	0.9994
M8	0.9986	0.9946
M9	0.8369	0.9875
M10	-0.2086	0.9849
M11	-0.0273	0.9760
M12	0.6263	0.9938

Table 3.2: Similarity metrics between reflectance of experimental and given values.

The results obtained in this study are not as consistent as those observed in the previous semester, where the spectral trends showed a much closer match. While several meteorite slices, such as M6 (represented in Figure 3.8) and M8, present a very strong agreement with both high Pearson and cosine similarity values, others display more moderate correlations.

In particular, the last three samples (M10, M11 and M12) exhibit the weakest agreement. M10 (represented in Figure 3.9) and M11 show negative or near zero Pearson correlation coefficients, indicating that the overall spectral trends are poorly reproduced or even inversely related. Their cosine similarity, however, remains high. This suggests that although the general shape of the curves may still be similar in magnitude, the relative variations across wavelengths are not well captured. M12 shows a moderate Pearson correlation, which indicates a partial agreement but still noticeably deviated from the reference data. In ??, the disagreement of M10 can be observed.

To study the influence of the averaging window size, a meteorite with promising results was selected. A representative point was chosen in a relatively homogeneous (bright) region of the sample, and averaging windows of 1×1 , 15×15 , 50×50 , and 100×100 pixels were analysed. This is represented in Figure 3.10, where the different window sizes are referred to as ROI in the legend of the figure.

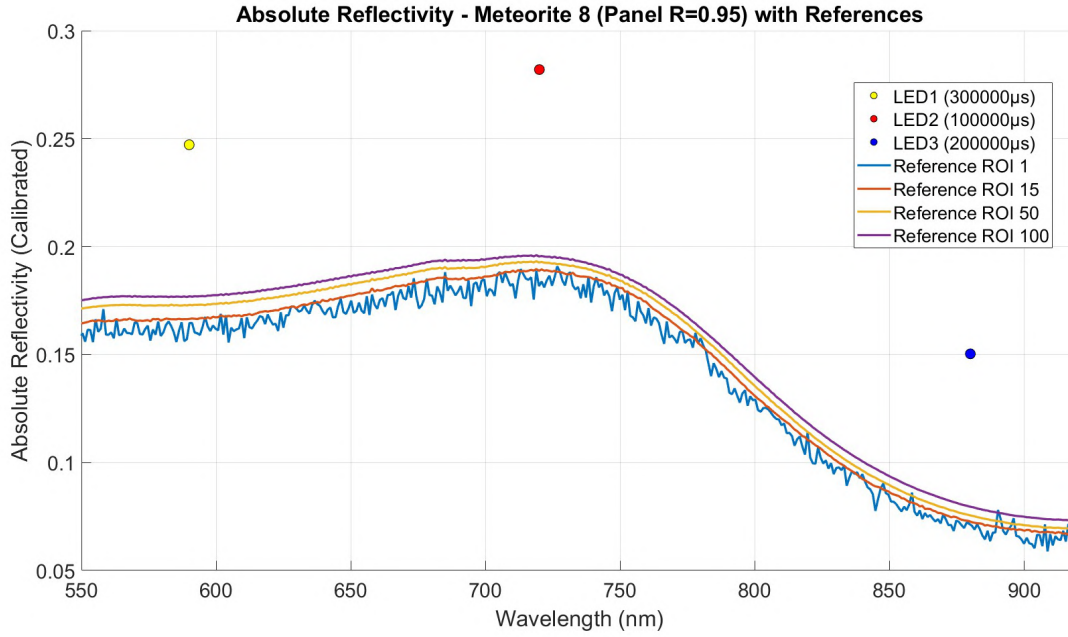


Figure 3.10: Comparison between measured data and reference data from average windows in spectral cube.

The first observation is that the overall spectral trend does not vary significantly with increasing window size. This suggests that the spectral cube contains a high level of redundancy and spatial information, and that moderate changes in the averaging window do not strongly affect the general behaviour of the measured reflectance. However, increasing the window size does improve the agreement in absolute reflectance values, as larger regions yield measurements that are closer to the true reflectance of the sample. It also indicates that the discrepancy between the averaging windows used in this work and those of the reference data is not the primary source of error for this meteorite.

However, it is important to note that this result depends on the specific region selected. In this case, the analysed area was relatively smooth and homogeneous in texture. For other meteorites, increasing the window size may introduce additional surface features, such as compositional variations, which are not present in the smaller regions analysed in the laboratory measurements. This can lead to inconsistencies.

Furthermore, the surface texture of the meteorites plays a significant role in the measurements. Meteorites often exhibit heterogeneous compositions, including metallic, crystalline, or glassy components, which can strongly influence the way light is reflected. Even small surface irregularities could alter the measurement, affecting the angle of reflection and consequently introducing variability in the measured brightness.

Therefore, the choice of a 15×15 window represents a compromise between reducing noise and avoiding the inclusion of unwanted features. Nevertheless, a more systematic approach for selecting the optimal average region could further improve the agreement between our experimental measurements and the reference data, shown in Table 3.2.

Several additional factors may also contribute to the observed discrepancies. The reference data

were not acquired under the same conditions or at the same time as the measurements in this work. The difference in illumination, calibration procedures, and acquisition setup are expected to introduce inconsistencies. Moreover, the reference spectral cube provides significantly higher spectral and spatial resolution. While hyperspectral data captures detailed features across many wavelengths, our system relies on only three wavelengths, which limits its ability to capture important features. Additionally, the spatial regions are not strictly equivalent, since the spectral cube data average is taken over a larger and more detailed pixel set. Even if both regions were selected proportionally, their exact correspondence cannot be guaranteed.

3.5.3 Experimental Results on Geological Samples: Rock Analogues

The objective of this part of the study was to evaluate whether the proposed method can be used to identify minerals from reflectance measurements obtained under LED illumination.

As discussed in subsection 1.1.2, olivine and pyroxene are minerals typically formed through igneous processes and are commonly found in planetary crusts and mantles. Their detection is therefore of particular importance, as their presence on Phobos would support a formation scenario involving material originating from Mars, rather than a captured primitive asteroid. Consequently, identifying these minerals provides valuable insight into one of the key scientific questions of the mission: the origin of Phobos and Deimos.

To achieve this, two analogue rock samples were analysed, each containing clearly visible crystals: a green olivine crystal and a black pyroxene crystal, shown in Figure 3.11. These crystals allowed specific regions corresponding to each mineral to be selected for analysis.



Figure 3.11: Green Olivine Crystal (on the right) and Black Pyroxene Crystal (on the left)

To assess the validity of the measurements, two types of reference spectra were used. First, spectral data of the same rock samples provided by CRPG were used as direct reference. These measurements were acquired on the same crystals analysed in this work, however, they were obtained independently and the exact location within the crystal is not known. As a consequence, a second reference was introduced using the general spectra of olivine and pyroxene from the U.S. Geological Survey (USGS) spectral library [34] were used. These represent

idealized spectral signatures of the minerals and allow comparison with the expected spectral behaviour of these materials.

During the experimental phase, an important limitation related to the illumination conditions was identified. The influence of sample height on LED illumination became evident, as the non-uniform illumination causes the halo patterns produced by the LEDs to shift with altitude. As a result, the calibration obtained for the meteorites was no longer applicable, and height dependent calibration became necessary.

To address this, images of the calibration target were acquired at three different altitudes. Intermediate values were obtained through interpolation, while values outside this range required extrapolation. However, this calibration remains approximate, as the profilometer measurements were not consistently reliable due to intermittent disconnections. Consequently, the height of the samples had to be estimated, with the olivine crystal assumed to be at 15 cm and the pyroxene crystal at 13 cm.

Olivine Crystal Tests

In Figure 3.12, three different datasets are shown: the measured data obtained in this work, represented by the three discrete points, the spectral profile of the same rock sample provided by CRPG, shown in green, and the general spectral profile of olivine from the USGS spectral library, shown in black.

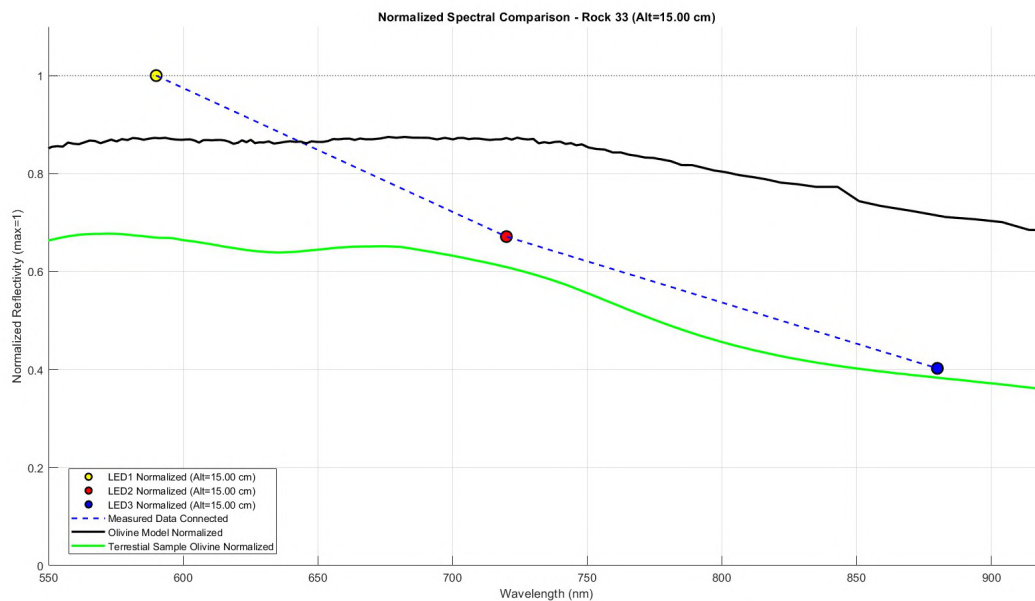


Figure 3.12: Olivine crystal comparison of Point 1 with reference spectral profiles.

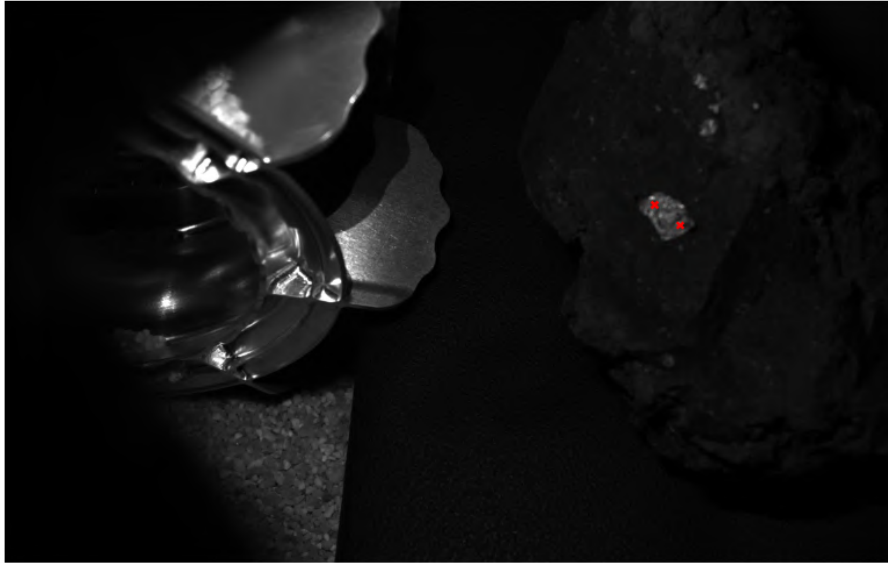


Figure 3.13: Two points chosen inside the crystal, illustrating the mentioned Point 1 (on the left) and Point 2 (on the right)

The quantitative results of this comparison are presented in Table 3.3 for two different locations on the same olivine crystal, referred to as Point 1 and Point 2 (see Figure 3.13). These points correspond to two distinct regions selected within the crystal in the acquired image.

Point	Reference	Pearson	Cosine
Point 1	Olivine Model (USGS)	0.8361	0.9642
	Terrestrial Sample Olivine (CRPG)	0.9284	0.9875
Point 2	Olivine Model (USGS)	0.5579	0.8024
	Terrestrial Sample Olivine (CRPG)	0.7118	0.8529

Table 3.3: Similarity metrics between measured reflectance and olivine reference spectra for two different points within the same crystal.

For Point 1, a strong agreement is observed, particularly with the CRPG reference. However, when analysing Point 2, located in a different region of the same crystal, the agreement decreases significantly for both references. This demonstrates that the measured reflectance is highly sensitive to the exact location selected, even within a single crystal.

It can also be observed that, in both cases, the agreement is consistently higher with the spectral profile provided by CRPG than with the general olivine model from the USGS library. This indicates that the measured data better captures the specific characteristics of the analysed sample rather than the idealized behaviour of olivine. However, as mentioned previously, this agreement should be interpreted with caution, since the exact location of the spectral measurement in the CRPG reference is unknown. Moreover, the measurements were acquired under different experimental conditions, which introduced uncertainty and may contribute to the observed discrepancies.

Generally, the observed variability between the points can be attributed to several factors.

Natural samples are inherently heterogeneous, meaning that even within a single crystal there may be variations in composition or internal structure. In addition, surface effects play an important role, as small changes in surface roughness or orientation can modify the way light is reflected. Then of course, our experiments are all limited by spatial resolution, and the region selected could include mixtures of different mineral phases or boundaries.

Pyroxene Crystal Tests

In Figure 3.14, three different datasets are presented: the measured data obtained in this work, represented by the three discrete points; the spectral profile of the same rock sample provided by CRPG, shown in green; and the general spectral profile of pyroxene from the USGS spectral library, shown in black. The quantitative results for two different points are reported in Table 3.4.

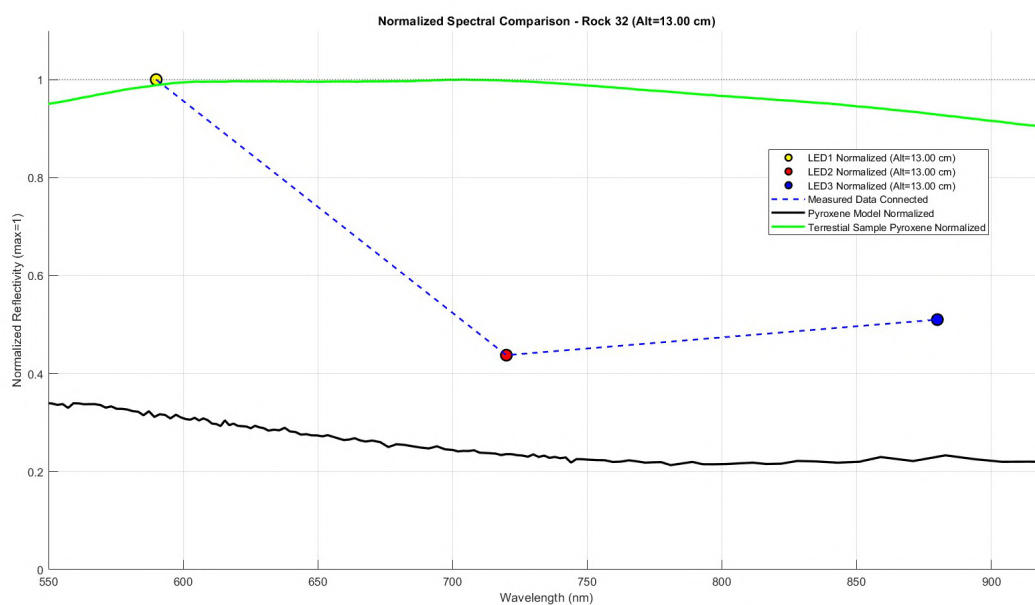


Figure 3.14: Pyroxene crystal comparison of Point 1 with reference spectral profiles.

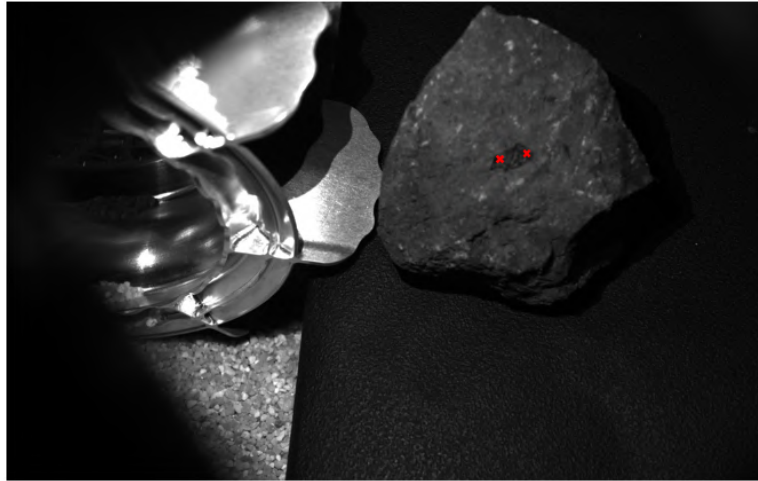


Figure 3.15: Two points chosen inside the crystal, illustrating the mentioned Point 1 (on the left) and Point 2 (on the right)

Point	Reference	Pearson	Cosine
Point 1	Pyroxene Model (USGS)	0.9495	0.9918
	Terrestrial Sample Pyroxene (CRPG)	0.1413	0.9675
Point 2	Pyroxene Model (USGS)	0.9913	0.9473
	Terrestrial Sample Pyroxene (CRPG)	0.3230	0.8960

Table 3.4: Similarity metrics between measured reflectance and pyroxene reference spectra for two different points within the same sample.

A similar behaviour to that observed for olivine can be seen, as the results vary significantly depending on the selected point within the same crystal (see Figure 3.15 for reference). For instance, as shown in Table 3.4, the Pearson correlation for the CRPG reference increases from 0.1413 at Point 1 to 0.3230 at Point 2, while the cosine similarity also varies. This confirms the measured reflectance is very sensitive to the exact location selected.

However, in contrast to the olivine results, the agreement with the general pyroxene spectral model from the USGS library is consistently higher. In particular, Pearson correlation values of 0.9495 and 0.9913 are obtained for Points 1 and 2, respectively, which are significantly higher than those obtained with the CRPG reference. A similar trend is observed in the cosine similarity values.

This suggests that the measured data more closely follow the characteristic spectral behaviour of pyroxene as represented in the USGS model, while the lower agreement with the CRPG reference may be due to additional variability in the sample such as surface conditions or measurement differences.

Updated Testbed Configuration Tests

Before additional measurements could be performed, the testbed was updated as described in subsection 1.1.4. As a result, the visual appearance of the rock changed due to modifications in the camera optics, viewing geometry, and illumination conditions. This difference can be observed in Figure 3.16 and 3.17, where the same sample is shown under both configurations.



Figure 3.16: Basalt rock with pyroxene crystal in previous configuration.



Figure 3.17: Basalt rock with pyroxene crystal in updated configuration.

Once again, in Figure 3.18, three different datasets are presented: the measured data obtained in this work, represented by the three discrete points, the spectral profile of the same rock sample provided by CRPG, shown in green, and the general spectral profile of pyroxene from the USGS spectral library, shown in black. The quantitative results are summarized in Table 3.5.

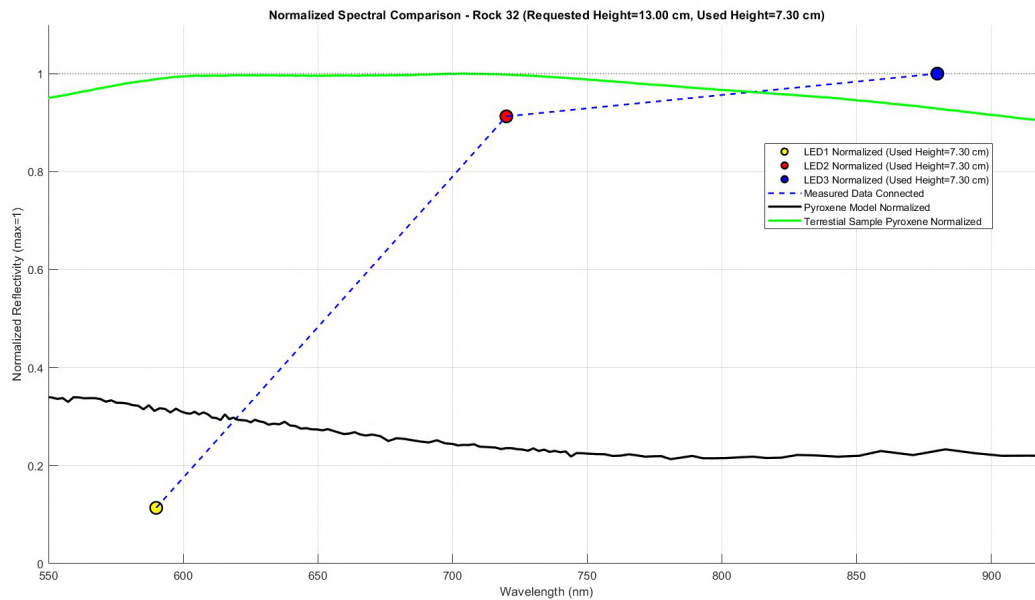


Figure 3.18: Pyroxene crystal comparison with spectral profiles under new configuration

Compared to previous results, a clear degradation in the correspondence between the measured

Reference	Pearson	Cosine
Pyroxene Model (USGS)	-0.9995	0.7785
Terrestrial Sample Pyroxene (CRPG)	-0.4725	0.8531

Table 3.5: Similarity metrics between measured reflectance and pyroxene reference spectra for an additional selected point.

reflectance and the reference spectra is observed, particularly in the Pearson correlation values. These values become negative for both references, as seen in Table 3.5, indicating that the measured trend is effectively inverted with respect to the reference spectra.

At the same time, the cosine similarity values remain relatively high, despite the clear visual mismatch between the curves. This suggests that cosine similarity should be interpreted with caution in this context, as it is less sensitive to differences in the shape of the spectra and can yield high values even when the spectral trends differ. This behaviour is particularly evident when using a limited number of wavelengths, where the metric may not fully capture the discrepancies in the spectral trend. Nevertheless, it remains useful as a complementary measure of overall signal similarity. Further investigation could be carried out to identify more suitable metrics that better capture differences in spectral behaviour for this type of data.

One possible explanation for the observed discrepancy is the change in viewing geometry introduced in the updated testbed configuration. Modifications to the camera angle alter the way reflected light is captured and, therefore, affect the measured reflectance. The influence of incidence and observation angles on reflectance measurements is well known in reflectance spectroscopy, however, in this setup, these geometric factors were not controlled or characterised in detail, which may explain the degradation in agreement.

In fact, an improvement in the results was initially expected, as the LEDs were better aligned in the updated configuration. The fact that this improvement is not observed suggests that additional factors are influencing the measurements, most likely related to surface interaction effects or variations in the selected regions.

An issue was identified in the calibration model related to the integration time and the height-dependent correction. As a result, the current height model does not behave reliably when extrapolating beyond the available calibration range. In practice, the model defaults to using the highest available calibration height. It should be noted that this maximum height is still relatively low and does not fully cover the expected range of measurements. Due to the updated camera configuration, it was not possible to acquire additional calibration data at larger heights. For this reason, the calibration procedure should be repeated in future work, following the recommendations outlined in subsection 3.5.4, in order to obtain a more consistent height model.

Despite the degradation in similarity, this test provides useful insight into the sensitivity of the measurements to geometric and calibration related factors. This highlights the importance of carefully controlling acquisition conditions and suggests that further investigation of these effects would be beneficial for improving the reliability of our method.

3.5.4 Future Work and Experimental Guidelines

Future Work

Based on the limitations identified in this study, several directions for future work are proposed. First, it would be useful to return to experiments using a spectrometer. At this stage, having reliable and reproducible measurements is more critical than testing many different materials. While identifying minerals is valuable, the differences in experimental conditions make it difficult to draw clear conclusions. Establishing a consistent way of taking measurements would allow for a better assessment of the system's capabilities. In this context, it would also be useful to study how different sample textures affect the measurements. Simpler and more controlled materials, such as powdered samples, could help separate the effect of composition from the effect of surface texture.

Another important point is the effect of geometry. Further tests should be carried out to understand how the measured reflectance changes with the position of the sample relative to both the LEDs and the camera. Simple and controlled experiments could help determine how sensitive the system is to these changes, and possibly identify a more stable configuration.

In this context, a detailed characterization of the LEDs in three-dimensional space would be very useful. Using the calibration panel, it should be possible to estimate the intensity distribution of each LED as a function of position. This would help account for the ring patterns observed in the images, which indicate that the illumination is not uniform across the scene and would significantly improve the accuracy of the reflectance estimation.

Another important improvement concerns the use of the profilometer. Ensuring that it provides consistent and reliable measurements should be a priority. Once this is achieved, the current height model can be refined and extended so that distances are defined relative to the LEDs rather than to the ground, which would result in a more physically meaningful calibration. This aspect would work together with the detailed LED characterization.

An additional direction for future work would be to focus on the study of surface reflectance and possible albedo variations, rather than on direct mineral identification. Given the limited spectral resolution of the current setup and the strong influence of geometry on the measurements, it may be more realistic to first assess whether the system can provide reliable and reproducible information about how much light different surfaces reflect. In this context, future experiments could investigate whether the setup is able to distinguish between darker and brighter analogue materials under controlled conditions, and whether these differences remain stable when the viewing geometry changes. This would provide a more robust basis for the interpretation of measurements acquired on the surface of Phobos.

Finally, in the longer term, it would be interesting to study analogue materials representative of the surface of Phobos. The goal would be to understand what kind of spectral response could be expected, allowing for more informed interpretation of in-situ measurements. However, this represents a more complex challenge and should be addressed once the experimental setup and calibration procedures are fully validated.

Experimental Guidelines

To ensure consistency and reproducibility, future experiments should follow a structured protocol:

- **Pre-experiment planning**

- Define the objective of the test (e.g., calibration, sample measurement, geometry study).
- Select the region of interest on the sample and ensure it can be consistently identified.
- Record the integration time, LED used, and sample for each measurement.

- **Profilometer usage**

- Measure and verify the zero point before starting the experiment.
- Always measure and record the height of the sample before taking images.
- Ensure that profilometer readings are stable and repeatable.

- **Calibration measurements**

- Perform calibration measurements before any sample measurements.
- Acquire data at multiple heights covering the full working range.
- Record the height using the profilometer for each calibration point.

- **Sample measurements**

- Acquire images of each sample using all available LEDs.
- Maintain the same region of interest across all measurements.
- Verify that all data is correctly saved before moving to the next measurement.

- **Post-processing**

- Apply calibration, normalization, and comparison steps consistently.
- Verify results for anomalies before drawing conclusions.

For the more complex experiments discussed in subsection 3.5.4, the following recommendations are proposed:

- **LED characterization**

- Use the calibration panel to acquire measurements at different heights, covering a wide range of distances.
- Cover as much of the field of view as possible with the calibration panel to map illumination variations. If necessary, move the calibration panel at the same height to increase the number of pixels covered.

- **Angle study**

- Once the LED characterization is complete, perform measurements on a smooth sample (i.e., a surface without strong reflective features such as metallic or crystalline components).
- Use a spectrometer as a reference measurement.
- Rotate the sample to different angles with respect to both the LEDs and the camera, including perpendicular configurations and intermediate positions.
- Measure and record the height using the profilometer for each configuration, as this is required to estimate the LED intensity.
- Analyse how the change in angle affects the measured reflectance.

4 Conclusion

In this research, we contributed to the scientific preparation for the MMX mission by developing image processing tools that support the analysis of WheelCam images from the IDEFIX rover. Specifically, we focused on advancing two core objectives: estimating wheel sinkage and exploring the potential for spectral analysis using panchromatic images illuminated by narrow-band LEDs. To achieve this, we improved existing algorithms for wheel detection and ellipse fitting, enabling more accurate identification of the wheel-regolith interface across varying conditions. Additionally, we continued to develop the previous framework to assess whether spectral information about the regolith could be inferred from images taken under different illumination wavelengths. These tasks were facilitated by using a dedicated testbed designed to simulate the conditions on Phobos, allowing us to validate our methods in a controlled environment. Through this work, we laid essential groundwork for supporting key scientific goals of the MMX mission, particularly in understanding the mechanical and compositional properties of Phobos' regolith.

Wheel sinkage

The key conclusions regarding the sinkage estimation part of this research are summarized as follows. First, the wheel contour detection and ellipse fitting pipeline demonstrated reliable performance on rear WheelCam images acquired under several regolith simulants. The algorithm successfully identified the wheel contour for JAXA, Moon, and gravel simulants, indicating that the approach is robust to variations in regolith reflectivity and surface texture. Among these cases, the best performance was observed for the JAXA regolith simulant, which is considered the closest analogue to the expected surface conditions on Phobos.

For the front WheelCam, the ellipse fitting algorithm was evaluated under varying illumination conditions and regolith simulants. A strong dependence was observed between ellipse fitting accuracy, signal-to-noise ratio, and regolith reflectivity. In particular, the results show that an optimal saturation level exists and depends on the reflectivity of the regolith. For the JAXA simulant, the optimal saturation level was found to be between 8 around 13%, whereas for the more reflective Moon simulant the optimal value is approximately 3%. These results indicate that exposure time will need to be tuned during rover operations in order to ensure reliable wheel detection and contour estimation.

The complete sinkage estimation pipeline was tested using measurements from the vertical laser in the experimental testbed. In the validation experiment, the automatic pipeline estimated the wheel sinkage with an absolute error of 4.6% of the wheel radius relative to the laser measurement. While this result demonstrates the feasibility of automatic sinkage estimation from WheelCam imagery, the validation dataset remains limited and the laser measurements used as reference are subject to uncertainties.

The validation experiments conducted with the revised testbed setup also revealed an important limitation of the current method. When only a small portion of the wheel contour is visible in

the image, the ellipse parameters become poorly constrained, leading to significant uncertainty in the predicted wheel geometry. This limitation was observed with the updated and more representative camera configuration in the testbed, where the visible portion of the wheel is reduced compared to the previous setup.

The next major steps needed to progress this research are presented in the following task breakdown:

1. Optimize the contour detection and/or ellipse fitting algorithms for the new front Wheel-Cam viewing angle in the testbed
2. Expand the sinkage validation dataset to include a wider range of sinkage levels and regolith types
3. Automate the wheel-regolith interface detection using a regolith segmentation model, enabling the sinkage estimation pipeline to be fully autonomous

Spectral studies

This study has explored the possibility of detecting material composition using the IDEFIX rover panchromatic cameras under controlled LED illumination, extending the simplified work of the previous semester toward solid samples such as meteorites and analogue materials. The introduction of a calibration panel made it possible to move from brightness trends to estimated reflectance values, which allowed for a more meaningful comparison with the reference data.

Several of the results obtained in this study were promising. For some meteorites, a strong agreement with the reference was observed and for the analogue samples, certain selected points within olivine and pyroxene crystals showed similarities with the expected spectral profiles. These results indicate that the measured data does contain physically meaningful information and that the combination of the LEDs and the panchromatic camera is sensitive to differences in surface composition and texture.

However, several limitations were also identified. The comparison data was not always acquired under the same experimental conditions, and in the case of the analogue materials, the exact spatial point associated with the reference spectrum was unknown. Moreover, the results are shown to vary significantly depending on the selected point, the averaging window, the surface texture and the positioning angle. In particular, the updated testbed configuration highlighted how strongly the measured reflectance can change when the camera and illumination angle are modified.

The calibration process also introduced important uncertainties. The height correction could not be modelled reliably due to the limited calibration data and because of the unstable profilometer measurements. As a result, extrapolation is not reliable, and the reflectance values for the analogue samples should be considered approximate. This limits the confidence of our results.

For this reason, although the results are not yet sufficiently reliable to draw clear conclusions about mineralogical identification, the method cannot be dismissed at this stage. Several

aspects of the approach are not yet fully controlled, including calibration, angle, and sample complexity, and further work is required before a definitive assessment can be made.

Therefore, the main outcome of this work is that the proposed approach is feasible, but still requires validation before scientific conclusions can be made. IN particular, more experiments are still needed to understand the role of geometry, sample texture, LED illumination patterns, and the dependency on height of the calibration. Once these factors are better characterized, it will be possible to assess whether the system can contribute to mineralogical identification or, more realistically, to the study of reflectance and possible albedo variations on the surface of Phobos.

The next steps, detailed in subsection 3.5.4, should therefore focus on the following:

1. Return to controlled experiments with a spectrometer in order to obtain reproducible reference measurements under the same conditions.
2. Study the influence of sample texture and geometry in a systematic way, including the relative position of the sample with respect to both the LEDs and the camera.
3. Improve the calibration procedure by characterizing the LEDs in three-dimensional space and by obtaining more reliable height measurements with the profilometer.
4. In the longer term, investigate analogue materials representative of the surface of Phobos, with the aim of understanding reflectance behaviour and possible albedo variations under controlled conditions.

A Meteorite Samples

M3: Lunar Meteorite NEA 078



Figure A.1: M3: Lunar Meteorite NEA 078

M5: Pallasite Seymchan



Figure A.2: M5: Pallasite Seymchan

M6: Ordinary Chondrite NWA 7938



Figure A.3: M6: Ordinary Chondrite NWA 7938

M8: Ordinary Chondrite Taoussat-n-Amouzzour 001



Figure A.4: M8: Ordinary Chondrite Taoussat-n-Amouzzour 001

M9: Howardite NWA 14804

Figure A.5: M9: Howardite NWA 14804

M10: Carbonaceous Chondrite NWA 12651

Figure A.6: M10: Carbonaceous Chondrite NWA 12651

M11: Carbonaceous Chondrite CV-CK



Figure A.7: M11: Carbonaceous Chondrite CV-CK

M12: Carbonaceous Chondrite Allende (CV3)

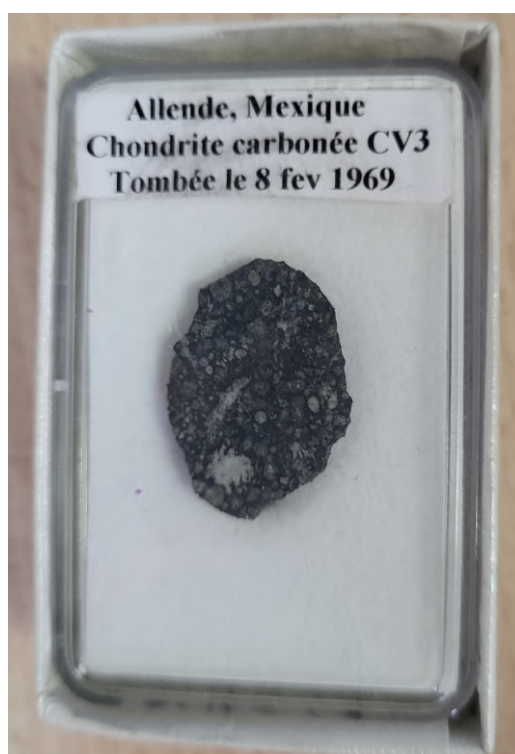


Figure A.8: M12: Carbonaceous Chondrite Allende (CV3)

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